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# A Method for Disentangling Multiple Treatments from a Regression Discontinuity Design

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**ABSTRACT**

In many settings, a policy discontinuity comprises several treatments that cannot be separately identified using a standard regression discontinuity design. I propose a method for identifying distinct treatment components from a single discontinuity by exploiting the asymmetry between entities entering versus exiting treatment. Using data from New York City for 2009-2013, I apply my strategy to the discontinuity associated with the introduction of class size caps – a widespread approach for reducing class sizes. I find that class size reductions increase student achievement, although these gains are counteracted by a newly-hired teacher effect. The method has broad potential applicability.

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# 1 Introduction

In a host of settings, policy discontinuities provide a credible means of identifying the effects of policies based on regression discontinuity (RD) designs. These designs have been used extensively as a result, reflected in burgeoning numbers of applications by empirical researchers (see the comprehensive survey by Lee and Lemieux (2010), for example). Often, a given discontinuity is composed of several individual treatments, each potentially relevant to academics and policymakers. Yet standard RD designs are only able to identify the *combined* effect of the treatment components that change at the threshold, which hinders more finely grained research and policy analysis.

In response to this limitation, I propose a new methodology that allows researchers to disentangle individual treatment effects from a discontinuity incorporating multiple treatment components. In essence, it exploits an asymmetry between entities entering versus exiting treatment. The approach can be used in any RD application where three main requirements are met: (i) panel data are available, (ii) entities receive shocks to their forcing variable (causing treatment variation over time), and (iii) one treatment component only occurs when entities enter or exit treatment. As many RD applications meet these necessary requirements, the approach has broad potential applicability, as the range of examples discussed at the end of the Introduction suggests.

One notable instance where a policy discontinuity is composed of multiple treatment components arises when studying class size caps using the RD design. The goal of RD studies in this area has been to estimate the impact of class size on student achievement using observational data. The RD studies typically find small effects,[1] in contrast with the much larger findings from experimental studies.[2] One possible reason for the differing results is that, unlike randomized experiments that assign students *and* teachers randomly to large or small classes, observational studies do not measure the 'pure' effect of class size on student achievement. Rather, they combine two treatment components bundled together under class size reduction: the impact of small class sizes, and the effect of possible changes in teacher quality due to the need to hire an additional teacher; indeed, several papers employing RD methods find significant discontinuities in observable teacher covariates around the class size

[1]Regression discontinuity designs show mixed results: some papers conclude that smaller class sizes improve student test scores (notably Angrist and Lavy, 1999; Urquiola, 2006; Fredriksson, Öckert, and Oosterbeek, 2013), while others find no significant effects (Hoxby, 2000; Angrist, Battistin, and Vuri, 2017; Angrist, Lavy, Leder-Luis, and Shany, forthcoming). I provide an overview of RD papers and other quasi-experimental studies in Appendix B.

[2]Using the Tennessee Project STAR experiment, both Finn and Achilles (1990) and Krueger (1999) conclude that smaller class sizes led to large improvements in student test scores. Subsequent analyses also show that Project STAR improved longer term outcomes (see Krueger and Whitmore (2001) and Chetty, Friedman, Hilger, Saez, Schanzenbach, and Yagan (2010)).



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cap threshold.[3]

I develop the new methodology in this important education setting where the three requirements are satisfied. Using my approach to examine the impact of class size caps – a widespread and popular means of reducing class sizes that has seen abundant use of RD designs for estimation – allows me to recover the pure effect of class size for the first time using observational data, alongside a 'newly-hired' teacher effect.

To explain the methodology, suppose there is a maximum class size rule (as in the empirical application below) stating that no more than thirty-two students are allowed in a class, and consider two cases. In the first, a school *enters* treatment – for instance, where a school-grade with prior year enrollment of thirty-two students gains another student in the current year. Here, to comply with the class size cap, the school-grade must add an extra class, which entails hiring a new teacher. In the second case, a school *exits* treatment – when a school-grade with prior year enrollment of thirty-three experiences a decline in enrollment to thirty-two, for example. This prompts a class to be removed and necessitates the release of a teacher. Taken together, while class size changes in both cases, there is an asymmetry in that some schools have a newly-hired teacher[4] (the 'entering treatment' case) while others do not (the 'exiting treatment' case).

My identification strategy takes advantage of this asymmetry by using two RD designs to estimate the effects of adding a class and removing a class separately. Since the 'add a class' case requires a newly-hired teacher, the joint effect of reducing class size and hiring an additional teacher in producing student achievement is identified in this instance. In contrast, the 'remove a class' case involves no newly-hired teacher and therefore captures the effect of reducing class size without an analogous newly-hired teacher effect.

The pure effect of class size is point-identified in this second case if teachers are released randomly, since teacher quality is then unchanged (on average) when a school removes a class. This implies that teacher quality is the same for school-grades that just do not remove a class and those that do, allowing the 'remove a class' case to uniquely capture the pure class size effect. If teachers are released non-randomly, then the RD estimate in the 'remove a class' case provides a lower bound to the pure class size effect (and this works *against* discovering a positive pure class size effect).[5] The RD estimator in the 'remove a class'

[3]See, for instance, the discontinuity in teacher education around the threshold noted in Fredriksson et al. (2013) (see Table IV).

[4]Throughout the paper, I refer to a 'newly-hired' teacher as someone who is new to the school-grade. This categorization may include some teachers who are assigned from another grade in the same school. Research has shown that newly-hired teachers improve with general experience (Rockoff, 2004), in-service training (Angrist and Lavy, 2001), and grade-specific experience (Ost, 2014), implying that, on average, newly-hired and newly-assigned teachers are likely to be of lower-than-average quality in the short run.

[5]In my application principals have discretion over personnel decisions and so (on average) lower quality




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case thus provides a conservative measure of the pure class size effect, while the difference between the two cases isolates the newly-hired teacher effect.

I apply my identification approach to the largest school district in the United States, New York City, for the 2009-13 period. Utilizing an administrative rule whereby no more than thirty-two students can be in a class, I show that smaller classes significantly improve student test scores: a four-student decrease in average class size at the school-grade level leads to an increase of about one-fifth of a standard deviation in mathematics scores and one-seventh of a standard deviation in English scores. These estimates are similar in magnitude to those reported by randomized experiments such as Project STAR. In practice, much of this effect is offset by the impact of the newly-hired teacher, who on average causes a decline in mathematics scores of about one-fifth of a standard deviation and of about one-sixth of a standard deviation in English at the school-grade level.

Overall, the results indicate that smaller class sizes do improve student achievement. Policymakers and school administrators need to be mindful, however, that these gains can be offset by changes in teacher quality, even in the absence of large-scale general equilibrium effects of the type highlighted clearly in Jepsen and Rivkin (2009).[6]

The wider applicability of the design proposed here relies on other RD applications satisfying the three requirements (noted above). In terms of requirements (i) and (ii), many RD applications meet the demands of panel data and year-to-year treatment variation (caused by shocks to entities' forcing variable). In education, for instance, treatment assignment is often determined *every* year based on whether student test scores or demographics exceed some threshold. Examples include: student assignment to gifted (Davis et al., 2013; Bui et al., 2014) or remedial education (Jacob and Lefgren, 2004; Schwerdt et al., 2017), teacher bonuses (Clotfelter et al., 2008; Dee and Wyckoff, 2015), and schools' accountability statuses (Figlio and Lucas, 2004; Rouse et al., 2013), program requirements (Chin et al., 2013) or funding (Van der Klaauw, 2008; Cowan and Goldhaber, 2018). These requirements are also

teachers are released, ensuring that the RD estimator provides a lower bound for the pure class size effect. To see this, note that the RD estimate in the 'remove a class' case is the difference in outcomes between school-grades that are affected and unaffected by the class size rule in the current period. The former are school-grades affected by the class size rule in both current and prior periods: such school-grades do not have a 'teacher release' effect. Those are compared with school-grades affected by the class size rule last period but which are now untreated, having released a low-quality teacher at the start of the current period. Only the latter will show a change in achievement, combining the effects of larger classes in the current period (presumably lowering achievement) with better average teacher quality, which *offsets* some of the loss caused by larger classes. Section 3.4 sets out this argument in detail and also shows that non-random teacher releases attenuates the newly-hired teacher effect toward zero.

[6]Jepsen and Rivkin (2009) study California's class size reduction program and find that the vast scale of the reform triggered a major teacher shortage, forcing many schools to reduce their class sizes by hiring inexperienced and uncertified teachers. They conclude that the detrimental effect of these new teachers served to offset much of the benefit of class size reduction, at least in the short-run.




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often met in non-education contexts: municipal financing (Brollo et al., 2013; Litschig and Morrison, 2013) or elections rules (Hinnerich and Pettersson-Lidbom, 2014) based on city size, environmental standards based on pollution levels (Chay and Greenstone, 2003), or even ratings of restaurants or financial products (Anderson and Magruder, 2012; McDonald and Rietz, 2018).

To apply the methodology elsewhere, requirement (iii) – that there is an entry-exit asymmetry – must also hold. This asymmetry is likely to arise in three broad situations, relating to: (a) administrative rules, (b) fixed costs or habit formation, and (c) underlying preferences or technologies. In situation (a), administrative rules cause an additional treatment component to be applied when entities enter treatment, but not when they exit (or vice versa). An example is this paper's class size application where schools hire a new teacher when entering treatment (i.e., add a class) but not when exiting treatment (i.e., remove a class). Another example is offered in Chiang (2009), where schools that just fail an accountability measure face sanction threats and receive a one-time funding boost. In that context, the policy causes entering schools to receive both sanction threats and a funding increase while exiting schools only lose the sanction threats. One could thus use the proposed methodology to uniquely identify the separate treatment effects of sanction threats and funding.

The entry-exit asymmetry arises in situation (b) because of habit formation or fixed costs. Consider, for instance, Litschig and Morrison (2013) where counties in Brazil received extra funding if their population size is above 10,188. Here, the authors show that increased public funding improves the education and income of citizens. A symmetric loss of funding, however, may not fully reverse the gains of increased public funding since higher funding in the past could improve current state capacity.[7] Related instances arise when entities entering treatment must make large fixed capital investments, which are unlikely to be unwound when they exit treatment (as with pollution abatement policies). For policy analysis, the extent of this asymmetry can help clarify whether policy benefits are likely to persist if a policy is ended.

Situation (c) occurs when an underlying preference or technology implies the existence of an entry-exit asymmetry. Take for example the pay tournament in Brehm et al. (2017), where teachers are paid a bonus if their students' test scores exceed some threshold. Under loss aversion,[8] we expect that individuals respond more to losing than gaining a bonus, and so any asymmetry in teacher responses when they lose (relative to gain) the bonus can identify the loss aversion parameter. Considering situations (a)-(c) together, all three requirements

[7]See, for instance, Dittmar and Meisenzahl (forthcoming) who show that increased public goods provision can change the political and institutional equilibrium.

[8]Using a randomized experiment, Fryer et al. (2012) showed that Chicago teachers exhibit significant loss aversion.




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are met in a variety of contexts, indicating that my proposed methodology can help applied researchers to disentangle the multiple components often embedded in RD designs with a view to performing more finely-grained policy analysis.

The rest of the paper is organized as follows: The next section introduces a model, showing which parameters are identified in randomized experiments and RD designs, respectively. This provides the basis for my empirical strategy, which is set out in Section 3. The empirical strategy also provides a correction for teacher turnover (which would otherwise cause the class size effect to be underestimated) and I discuss the implications of student tracking and differential class sizes in interpreting the class size parameter. Section 4 describes New York City's class size rule. The estimates are presented and placed in context in Section 5, and Section 6 concludes, drawing out the broader relevance of the methodology.

# 2 Model

To guide my econometric approach, I first set out the general model underlying the methodology. The model mimics a basic RD setup with two additional features: (i) there are two endogenous inputs, and (ii) one endogenous input depends upon the prior period's treatment status. This setup shows clearly how one can use the asymmetry of entities entering versus exiting treatment to identify both endogenous inputs. I then apply this model to a class size setting where the two endogenous inputs are class size and teacher quality. Here I demonstrate two main results: first, a regression discontinuity design using a maximum class rule will capture a combination of changes in class size and teacher quality. Second, these combined changes occur only for school-grades that add a class in response to the maximum class size rule, generating an asymmetry that allows for the effects of class size and teacher quality to be separately identified.

## 2.1 General Setup

The model consists of two periods: the current period, $t$, and the pre-period, $t-1$. The outcome for each entity $i$ at time $t$ is given by $Y_{it}$ and is enhanced by two inputs: $Z_{i1t}$ and $Z_{i2t}$. For expositional clarity, these inputs are binary so that $Z_{ijt} \in \{0, 1\}$ for $j = 1, 2$, although all results follow if inputs are continuous (as will be clear in the class size example that follows). Assuming that the production technology that converts inputs into outputs,



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$g(\cdot)$, is linear and additively separable,[9] the outcome of each entity $i$ at time $t$ is:

$$Y_{it} = g(Z_{i1t}, Z_{i2t}) = \alpha_1 Z_{i1t} + \alpha_2 Z_{i2t}\,. \tag{2.1}$$

The researcher is interested in the causal effect of each input on the outcome and so wants to identify the parameters $\alpha_1$ and $\alpha_2$. While these parameters can be identified through a randomized experiment that randomly assigns one input to individuals while *holding the other input constant*, researchers often only have access to quasi-experimental data.

One prominent quasi-experimental setting is the RD design whereby inputs are a deterministic function of another variable, $X_{it}$, which (following RD terminology) is labeled the 'forcing variable.' Specifically, the first input follows the usual RD setup, where this input takes on a value of one if the forcing variable exceeds some cutoff $c$:

$$Z_{i1t} = \begin{cases} 1 & \textit{if}\, X_{it} > c \\ 0 & \textit{if}\, X_{it} \leq c\,. \end{cases} \tag{2.2}$$

Contrary to the canonical RD case, however, I consider a second input which is also a deterministic function of whether the forcing variable is above or below the cutoff $c$. The key feature of the second input that is necessary to separately identify the two parameters is that it depends on both whether the forcing variable exceeds some cutoff $c$ at time $t$ *and* $t-1$.

To fix ideas, let the second input be received only if the cutoff is newly exceeded in period $t$ (i.e., entity 'enters' treatment):

$$Z_{i2t} = \begin{cases} 1 & \textit{if}\, X_{it} > c \;\&\; X_{i,t-1} \leq c \\ 0 & \textit{otherwise}\,. \end{cases} \tag{2.3}$$

While I have specified a particular binary form for the inputs here, the methodology is applicable so long as two general conditions are met: (i) one input is determined by whether the forcing variable exceeds some cutoff $c$ at time $t$, and (ii) the other input is determined based on whether the forcing variable exceeds some cutoff $c$ at time $t$ *and* $t-1$.

In RD designs, researchers compare the outcomes of entities just above the cutoff $c$ at time $t$ to those just below. With the above setup, however, if entities have exceeded the cutoff

[9] The linear and additive functional form is assumed for expositional clarity. All theoretical results follow from a more general non-additive production technology that allows interactions between the two inputs.



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for the first time, they receive both inputs and thus the usual RD comparison identifies:

$$\mathbb{E}\left[Y_{it}|X_{it} > c\right] - \mathbb{E}\left[Y_{it}|X_{it} \leq c\right] = \alpha_1 + \delta\alpha_2\,, \tag{2.4}$$

where $\delta$ is the proportion of entities who have newly exceeded the cutoff (i.e., entered treatment).

The key intuition for the proposed methodology is that both parameters can be identified by separating the usual RD comparison into two mutually exclusive cases: (i) the entity exceeded the cutoff in $t-1$, and (ii) the entity did not exceed the cutoff in $t-1$. On one hand, the first case has no entities that have newly exceeded the cutoff and so the RD comparison identifies:

$$\mathbb{E}\left[Y_{it}|X_{it} > c, X_{i,t-1} > c\right] - \mathbb{E}\left[Y_{it}|X_{it} \leq c, X_{i,t-1} > c\right] = \alpha_1 - 0 = \alpha_1\,, \tag{2.5}$$

where the first equality follows from substituting the conditions in equations (2.2) and (2.3) into equation (2.1), implying that there is no second input (i.e., $Z_{i2t} = 0$) in the rightmost term.

The second case, on the other hand, is such that all entities exceeding the cutoff have newly done so (i.e., they entered treatment) and so the RD comparison identifies:

$$\mathbb{E}\left[Y_{it}|X_{it} > c, X_{i,t-1} \leq c\right] - \mathbb{E}\left[Y_{it}|X_{it} \leq c, X_{i,t-1} \leq c\right] = (\alpha_1 + \alpha_2) - 0 = \alpha_1 + \alpha_2\,, \tag{2.6}$$

where the first equality follows because the conditions in equations (2.2) and (2.3) imply that the second input always occurs in the rightmost term (i.e., $Z_{i2t} = 1$). Since equations (2.5) and (2.6) form a system of two equations with two unknowns, both parameters can be identified through the two RD comparisons.

## 2.2 Class Size Application

I now apply the above general setup to a class size context where the two endogenous inputs are class size and teacher quality. This context also features some additional complexity as inputs are continuous and the running variable is discrete. While the identification argument is identical to that under the general setup, the adaptation to the class size context makes clear what drives the asymmetry in teacher quality between school-grades entering versus exiting treatment, which is key to the identification strategy to separate the effects of class size and teacher quality.

As above, the model consists of two periods: the current period, $t$, and the pre-period,



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$t-1$. Let $N_t$ denote school-grade enrollment at time $t$. Denote the average 'ability' of children in a given school grade at time $t$ by $\mu_t$, which consists of innate ability and all prior educational inputs the students have received up to the start of a school-grade; as such, it can grow over time.

Learning is also enhanced by the school inputs students receive during each period, namely class size, $c_t$, and teacher quality, $q_t$. Teacher quality is given by the 'type' of teacher in the school-grade, of which there are two: those currently serving the school-grade, whose quality is $q_{cur}$, and newly-hired teachers with quality $q_{new}$. I assume that the production technology that converts school inputs into outputs, $g(\cdot)$, is linear and additively separable. Thus, the average achievement, $A_t$, of the $N_t$ students in a school-grade at time $t$, which depends on ability and school-grade inputs, is:

$$A_t = g(c_t, q_t, \mu_t) = \alpha_c c_t + \alpha_q q_t + \mu_t \,. \tag{2.7}$$

This setup can be used to clarify the parameters identified through different empirical designs. First, consider a randomized experiment, like Project STAR, which assigns small and large classes randomly among teachers and students at time $t$. Accordingly, students assigned to small classes experience a decline in class size, while both student ability and the probability of being taught by a newly-hired teacher are identical (due to randomization) across the different class sizes. Teacher quality is thus held fixed and the experiment will capture:

$$\frac{\partial}{\partial c_t} g(c_t, q_t, \mu_t) = \alpha_c \,. \tag{2.8}$$

Throughout, I characterize the parameter $\alpha_c$ as measuring the 'pure class size' effect.[10]

Now, consider an RD design using a maximum class size rule. The rule compels school-grades with enrollment just above the class size cap to have an additional class, creating small classes in those school-grades, while those with enrollment just below the class size cap are not induced to create an additional class and thus have large classes. Letting $\gamma$ represent the number of classes, the average class size, $c_t$, of a school-grade is enrollment divided by

[10]Prior researchers have concluded that parental investments adjust to class size changes (BonesrØnning, 2004; Datar and Mason, 2008). In a richer model with parental inputs, one could separate out the class size effect operating within-school and through changes in parental investments. However, since parental inputs are typically unobserved, these two channels often cannot be distinguished. Therefore, as in randomized experiments such as Project STAR, the pure class size effect discussed in the paper incorporates parental investment responses also.



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number of classes, $\frac{N_t}{\gamma}$, with $\gamma$ set such that average class size is below the class size cap $\bar{c}$:

$$c_t = c(N_t) = max\left\{\frac{N_t}{\gamma} : \frac{N_t}{\gamma} \leq \bar{c}, \gamma \in \mathbb{N}\right\} . \tag{2.9}$$

Assuming that changes in school-grade enrollment are small enough that at most one class is added[11] and that newly-hired teachers are not disproportionately assigned large or small classes, teacher quality is given by:

$$q_t = q(N_{t-1}, N_t) = q_{cur} \cdot \mathbb{1}\left\{N_t \Big/ \frac{N_{t-1}}{c(N_{t-1})} \leq \bar{c}\right\} + \frac{\frac{N_{t-1}}{c(N_{t-1})} q_{cur} + q_{new}}{\frac{N_{t-1}}{c(N_{t-1})} + 1} \cdot \mathbb{1}\left\{N_t \Big/ \frac{N_{t-1}}{c(N_{t-1})} > \bar{c}\right\} . \tag{2.10}$$

Here, $\frac{N_{t-1}}{c(N_{t-1})}$ denotes the number of classes in the school-grade last year. Therefore, the term $\mathbb{1}\left\{N_t \Big/ \frac{N_{t-1}}{c(N_{t-1})} \leq \bar{c}\right\}$ takes value one when enrollment this period divided by the number of classes last period is below the class size cap. In that case, no new classes are added, and so teacher quality for the school-grade is given by $q_{cur}$. Similarly, $\mathbb{1}\left\{N_t \Big/ \frac{N_{t-1}}{c(N_{t-1})} > \bar{c}\right\}$ takes value one if current period enrollment divided by the number of classes last period rises above the class size cap $\bar{c}$, causing a new class to be added. In that case, one new teacher will be hired and so teacher quality will be given by $\frac{\frac{N_{t-1}}{c(N_{t-1})} q_{cur} + q_{new}}{\frac{N_{t-1}}{c(N_{t-1})} + 1}$ since the $\frac{N_{t-1}}{c(N_{t-1})}$ classes present last year will be taught by the same teachers while one class will be taught by a newly-hired teacher.

The RD design compares school-grades whose enrollment is just above or just below the class size cap. It therefore contrasts school-grades with an enrollment of $N_t$ and an average class size of $c(N_t) = \frac{N_t}{\gamma} = \bar{c}$ to those whose enrollment in period $t$ just exceeds the class size cap, so that $\frac{N_t}{\gamma} > \bar{c}$, necessitating the school-grade to have another class with an average class size of $c(N_t) = \frac{N_t}{\gamma+1} << \bar{c}$.

I now focus on school-grades just above and just below the class size cap in period $t-1$. At the start of period $t$, there is an enrollment shock and school-grades subsequently make personnel decisions based upon the number of classes they are required to have, which is determined by whether their enrollment is above or below the class size cap according to equation (2.9). For school-grades with period $t-1$ enrollment below the class size cap, a positive enrollment shock causes them to exceed the class size cap. A new class then gets added and we label this the 'add a class' case. Conversely, for school-grades whose

[11] This ensures that only *one* teacher is added. In practice, school-grades almost never have enrollment shifts large enough such that two classes are added.



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enrollment was above the class size cap in period $t-1$, a negative enrollment shock allows them to eliminate a class at time $t$ without violating the maximum class size rule. Here, these school-grades remove a class and are thus collectively labelled the 'remove a class' case. This maps to the empirical design where a single student entering or leaving a school-grade by random chance determines whether or not a school-grade is affected by the rule.

Following the logic of the general setup, the effect of the maximum class size rule can be examined separately in each case. In the 'add a class' case, the maximum class size rule only affects school-grades that receive a positive enrollment shock. The maximum class size rule compels these school-grades to reduce class sizes by adding another class, which entails hiring a new teacher. In this case, the RD design identifies that the effect of the maximum class size rule on student achievement at time $t$ is:

$$\begin{aligned}&\mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} > \bar{c}, \frac{N_{t-1}}{\gamma} = \bar{c}\right] - \mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} = \bar{c}, \frac{N_{t-1}}{\gamma} = \bar{c}\right] \\ &= g\left(\frac{N_t}{\gamma+1}, \frac{\frac{N_{t-1}}{c(N_{t-1})}q_{cur} + q_{new}}{\frac{N_{t-1}}{c(N_{t-1})}+1}, \mu_t\right) - g\left(\frac{N_t}{\gamma}, q_{cur}, \mu_t\right) = \alpha_c\tilde{c} + \alpha_q \frac{q_{new} - q_{cur}}{\frac{N_{t-1}}{c(N_{t-1})}+1}, \end{aligned} \tag{2.11}$$

where the second equality follows from the linear technology in equation (2.7), $\tilde{c}$ represents the change in class size around the class size cap threshold and is given by $-\frac{N_t}{\gamma^2+\gamma}$, and $\frac{q_{new}-q_{cur}}{\frac{N_{t-1}}{c(N_{t-1})}+1}$ represents the change in teacher quality caused by the newly-hired teacher.

For the 'remove a class' case, in contrast, school-grades that experience either no enrollment change or a positive enrollment shock are subject to the maximum class size rule. Here, no new teacher is required, although school-grades that are no longer affected by the class size rule due to a negative enrollment shock remove a teacher. Looking forward to the econometric methodology, school-grades that remove a class will act as a counterfactual for those that keep the class due to the class size rule. Since the counterfactual school-grades can remove a low-quality teacher, then this may offset some of the (possible) achievement loss caused by larger classes.[12] The RD design in this case identifies:

$$\begin{aligned}&\mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} > \bar{c}, \frac{N_{t-1}}{\gamma} > \bar{c}\right] - \mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} = \bar{c}, \frac{N_{t-1}}{\gamma} > \bar{c}\right] \\ &= g\left(\frac{N_t}{\gamma+1}, q_{cur}, \mu_t\right) - g\left(\frac{N_t}{\gamma}, q_{cur}, \mu_t\right) = \alpha_c\tilde{c}\,, \end{aligned} \tag{2.12}$$

where $\tilde{c} = -\frac{N_t}{\gamma^2+\gamma}$. The RD design in the setting therefore identifies the pure class size

[12]Section 3.4 specifies why, on average, it is lower-quality teachers who are released in my application (which is corroborated by the data). Further, it shows that, under such teacher releases, the class size effect estimated in this paper is a lower bound of the true class size effect.



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parameter, $\alpha_c$, scaled by the change in class size $\tilde{c}$.

The overall effect of the maximum class size rule, which combines effects from both the 'add a class' and 'remove a class' cases, is given by:

$$\begin{aligned}
&\mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} > \bar{c}\right] - \mathbb{E}\left[A_t \middle| \frac{N_t}{\gamma} = \bar{c}\right] \\
&= \delta \cdot g\left(\frac{N_t}{\gamma+1}, \frac{\frac{N_{t-1}}{c(N_{t-1})} q_{cur} + q_{new}}{\frac{N_{t-1}}{c(N_{t-1})} + 1}, \mu_t\right) + (1-\delta) \cdot g\left(\frac{N_t}{\gamma+1}, q_{cur}, \mu_t\right) - g\left(\frac{N_t}{\gamma}, q_{cur}, \mu_t\right) \\
&= \alpha_c \tilde{c} + \delta \cdot \alpha_q \cdot \frac{q_{new} - q_{cur}}{\frac{N_{t-1}}{c(N_{t-1})} + 1}, \qquad (2.13)
\end{aligned}$$

where $\delta$ denotes the proportion of school-grades whose enrollment was below the class size cap in $t-1$ (i.e., are in the 'add a class' case). Accordingly, using a maximum class size rule does not uncover the pure class size parameter from equation (2.8), $\alpha_c$, except in two special instances: first, the quality of newly-hired and current teachers is the same ($q_{new} = q_{cur}$), or second, there are no 'add a class' cases ($\delta = 0$). In the absence of those special circumstances, a researcher using a maximum class size rule recovers a weighted average of the pure class size parameter, $\alpha_c$, and the change in teacher quality. The researcher could, however, find the pure class size parameter from a maximum class size rule if the sample were limited to school-grades in the 'remove a class' case.

This discussion will guide the empirical analysis, where school-grades are partitioned into 'add a class' and 'remove a class' samples. The 'remove a class' sample identifies the pure class size parameter $\alpha_c$ (as shown in equation (2.12)). Estimates from the two samples are then differenced, subtracting equation (2.12) from equation (2.13), to recover the 'newly-hired teacher effect' at the school-grade level, given by $\alpha_q \frac{q_{new} - q_{cur}}{\frac{N_{t-1}}{c(N_{t-1})} + 1}$.

# 3 Econometric Design

Building on the intuition from Section 2, I develop an econometric framework to separately identify $\alpha_c$, the pure class size effect of equation (2.8), and the newly-hired teacher effect at the policy discontinuity generated by a maximum class size rule. The difficulty in separately identifying these effects stems from the fact that both class size and teacher quality are endogenously determined, and so an econometrician requires two sources of exogenous variation to identify them. The identification challenge is similar to that in Angrist and Lang (2004) who develop an empirical model with two endogenous inputs – class size and proportion of low-performing students – and then used exogenous variation provided by class size caps and school-grade-year variation to separately identify both effects on student



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achievement. I extend this idea to a regression discontinuity framework, using exogenous variation provided by school-grades entering *and* exiting the class size cap to separately identify the impact of class size and newly-hired teachers on student achievement.

## 3.1 Setup

I present my econometric design using the potential outcomes framework. To focus on the intuition underlying the design, I focus my discussion on the first class size cap around thirty-two students, although the design is easily extended to a multiple class size cap framework (see Appendix C). Furthermore, I assume that each school-grade follows the maximum class size rule perfectly[13] and hires an additional teacher whenever they add a class at time $t$ that was not present at time $t-1$.

**Potential Outcomes:** Let $C = \{1, 2\}$ denote the number of classes in a given school-grade, where $C = 1$ denotes one class (i.e., high class sizes) and $C = 2$ denotes two classes (i.e., small class sizes). I denote the number of teachers in the school-grade by $T$ and let the subscript $N$ indicate whether there is a teacher that is new to the school-grade. Therefore, around the class size cap of thirty-two, $T \in \{1, 2, 2_N\}$, where $T = 1$ denotes one teacher who was present in the school-grade in the prior year, $T = 2$ denotes two teachers in the school-grade, both of whom were present in the school-grade in the prior year, and $T = 2_N$ denotes two teachers in the school-grade, but where one of them is new to the school-grade while the other was present in the school-grade in the prior year. For now, I assume that there is no teacher turnover, so $1_N$ occurs with zero probability and school-grades with two newly-hired teachers also occur with zero probability. Section 3.4 relaxes this assumption.

Define the 'treated' as school-grades induced by the maximum class size cap to create a second class to reduce their average class size. Accordingly, $D = 1$ denotes school-grades that have low class sizes by having two classes ($C = 2$) and two teachers ($T \in \{2, 2_N\}$), while $D = 0$ denotes school-grades with high class sizes having one class ($C = 1$) and one teacher ($T = 1$). Treatment, $D$, is therefore given by:

$$D = \begin{cases} 1 \text{ if } C = 2,\ T \in \{2, 2_N\} \\ 0 \text{ if } C = 1,\ T = 1\,, \end{cases} \tag{3.1}$$

where the events ($C = 1$, $T \in \{2, 2_N\}$) along with ($C = 2$, $T \in \{1\}$) are zero probability since school-grades cannot have one class with two teachers or two classes with one teacher.

[13]Hence, this design identifies the local intent-to-treat (ITT) effect of the class size rule. As in an RD design, this design can be made 'fuzzy' by dividing the local ITT estimates by the probability a school-grade complies with treatment to obtain local treatment-on-the-treated effects.



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The main novelty in the econometric design is to exploit the fact that school-grades only add a newly-hired teacher in this context when they add a class that was not present in the prior period. Let, $Z \in \{1, 0\}$ denote treatment in the prior period, with $Z = 1$ denoting that the school-grade was 'treated' and thus had two classes in the previous year and $Z = 0$ denoting that the school-grade was 'untreated' and had one class in the prior year. Treated school-grades (i.e., $D=1$) can therefore be split into two cases: those that were treated last period (i.e., $Z=1$) and so have two teachers that were present last period and those that were untreated last period (i.e., $Z=0$) and so had to add a teacher that was not present last period to instruct the newly-created second class. Untreated school-grades (i.e., $D=0$) always have a teacher who was present in the prior period. The number of teachers and presence of a newly-hired teacher, $T$, is therefore a function of $D$ and $Z$ and is given by:

$$T(D, Z) = \begin{cases} 1 & \text{if } D = 0,\ Z \in \{0, 1\} \\ 2 & \text{if } D = 1,\ Z = 1 \\ 2_N & \text{if } D = 1,\ Z = 0\,. \end{cases} \tag{3.2}$$

Potential outcomes for school-grade $s$-$g$ at time $t$ are therefore $\{Y(d, t(d, z)) : d \in \{0, 1\},\ t \in \{1, 2, 2_N\},\ z \in \{0, 1\}\}$.

**Treatment Assignment:** In this context, treatment is a deterministic function of school-grade enrollment, $\gamma_{sgt}$. Let $X_{sgt}$ denote this treatment-determining variable (henceforth the 'forcing variable'), where $X_{sgt}$ is defined as school grade enrollment normalized by the class size cap, so that $X_{sgt} = \gamma_{sgt} - 32$. Treatment, $D_{sgt}$, for school-grade $s$-$g$ in period $t$ is thus given by:

$$D_{sgt} = \begin{cases} 1 \text{ if } X_{sgt} > 0 \\ 0 \text{ if } X_{sgt} \leq 0\,. \end{cases} \tag{3.3}$$

Similarly, treatment in the prior period, $Z_{sgt}$, for school-grade $s$-$g$ in period $t$ is:

$$Z_{sgt} = \begin{cases} 1 \text{ if } X_{sg,t-1} \leq 0 \\ 0 \text{ if } X_{sg,t-1} > 0\,. \end{cases} \tag{3.4}$$

**Parameters of Interest:** In this paper, I focus on two quantities of interest: (i) the 'pure' class size effect and (ii) the 'newly-hired' teacher effect. The 'pure' class size effect in this context is given by:

$$\beta_{CS} = \mathbb{E}\left[Y(1, 2) - Y(0, 1)\right]\,, \tag{3.5}$$

which evaluates the effect of lowering class size by removing a class and going from two



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teachers to one. Crucially, since there are no newly-hired teachers, there is no difference in teacher quality in $\beta_{CS}$. The newly-hired teacher effect is then given by:

$$\beta_{NT} = \mathbb{E}\left[Y(1, 2_N) - Y(1, 2)\right] , \tag{3.6}$$

which evaluates the effect of hiring a new teacher in contrast to having the same teacher from the prior year when there are two classes.

## 3.2 Regression Discontinuity Design

I now lay out the assumptions required to separately identify the class size effect from the newly-hired effect in this framework and discuss their implications and possible violations. I then show that the conventional RD estimand estimates a (weighted) combination of the pure class size effect and the newly-hired teacher effect.

**Assumptions:** Intuitively, the regression discontinuity design compares outcomes between school-grades that just exceed the class size cap and are thus treated to those just below the class size cap and are therefore untreated. For this comparison to be valid, I impose the following assumption:

**Assumption 1** *(Continuity of Potential Outcomes)*

$$\mathbb{E}[Y(1)|X_{sgt} = x] \textit{ and } \mathbb{E}[Y(0)|X_{sgt} = x] \textit{ are continuous in } x \textit{ at } x = 0 .$$

Assumption 1 is the usual RD continuity assumption that guarantees treatment, $D$, is as good as randomly assigned at $x = 0$ (Lee and Lemieux, 2010). This ensures that school-grades with values of the forcing variable just above and just below the class size cap are identical in the absence of treatment.

Over and above the usual RD assumption of continuity of potential outcomes, I also assume that conditioning on prior treatment is irrelevant *except* through changes in teacher quality induced by hiring a new teacher when a class is added. I divide the assumption into two parts to highlight different implications of the assumption:

**Assumption 2** *Potential Outcomes are Independent of $Z$ (except through $T$)*
*(i) Homogeneity*

$$\mathbb{E}[Y(1,2)|Z=1, X=0] = \mathbb{E}[Y(1,2)|X=0] \textit{ and } \mathbb{E}[Y(1,2_N)|Z=0, X=0] = \mathbb{E}[Y(1,2_N)|X=0]$$



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*(ii) No Teacher Release Effect:*

$$\mathbb{E}\left[Y(0,1)|Z=1,X=0\right]=\mathbb{E}\left[Y(0,1)|Z=0,X=0\right]\equiv\mathbb{E}\left[Y(0,1)|X=0\right]$$

Assumption 2(i) states that for school-grades with two classes, conditioning on past treatment status is irrelevant. A violation of this assumption would potentially occur if treated school-grades that were untreated last period (i.e., just added a class) differed on unobservable characteristics *when* $X=0$ relative to treated school-grades that were treated last period (i.e., remained at two classes) and those characteristics influenced outcomes. To assess the validity of this assumption, one could check for differences in observable characteristics between the $Z=0$ and $Z=1$ samples, which I do in Section 4.3. In the absence of large differences in observable characteristics, one would not expect large difference to arise between the potential outcomes, except through the newly-hired teacher effect.[14]

Assumption 2(ii) is the counterpart to Assumption 2(i) for school-grades with one class. In addition to homogeneity, however, Assumption 2(ii) implies that there cannot be a 'teacher release' effect since a teacher is released when $Z=1$, but not when $Z=0$. Intuitively, there would be no teacher release effect if teachers are released at random. Under non-random release of teachers, my methodology no longer point identifies the class size effect, but rather identifies its *lower* bound (see Section 3.4 for a detailed argument).

**Standard Regression Discontinuity Design:** Now consider the usual RD design, which, as I will show, estimates a (weighted) combination of the pure class size effect and the newly-hired teacher effect. In the standard RD design, the RD estimand compares outcomes between school-grades that just exceed the class size cap and are thus treated (i.e., $D=1$) to those who are just below the class size cap and are therefore untreated (i.e., $D=0$). Specifically, the RD estimand, $\tau_{SRD}$, yields:

$$\begin{aligned}
\tau_{SRD} &= \mathbb{E}\left[Y|D=1,Z=1,X=0\right]\cdot\mathbb{1}\{Z=1\}+\mathbb{E}\left[Y|D=1,Z=0,X=0]\right]\cdot\mathbb{1}\{Z=0\} \qquad (3.7)\\
&\quad -\mathbb{E}\left[Y|D=0,Z=1,X=0\right]\cdot\mathbb{1}\{Z=1\}-\mathbb{E}\left[Y|D=0,Z=0,X=0\right]\cdot\mathbb{1}\{Z=0\}\\
&= \mathbb{E}\left[Y(1,2)|Z=1,X=0\right]\cdot P(Z=1|D=1)+\mathbb{E}\left[Y(1,2_N)|Z=0,X=0\right]\cdot P(Z=0|D=1)\\
&\quad -\mathbb{E}\left[Y(0,1)|Z=1,X=0\right]\cdot P(Z=1|D=0)-\mathbb{E}\left[Y(0,1)|Z=0,X=0\right]\cdot P(Z=0|D=0)\\
&= \mathbb{E}\left[Y(1,2)|X=0\right]\cdot(1-\delta)+\mathbb{E}\left[Y(1,2_N)|X=0\right]\cdot\delta-\mathbb{E}\left[Y(0,1)|X=0\right]\\
&= \underbrace{\mathbb{E}\left[Y(1,2)-Y(0,1)|X=0\right]}_{\equiv\beta_{CS}}+\delta\cdot\underbrace{\mathbb{E}\left[Y(1,2_N)-Y(1,2)|X=0\right]}_{\equiv\beta_{NT}}\\
&= \beta_{CS}+\delta\cdot\beta_{NT}\,,
\end{aligned}$$

[14]To the extent there are unobservable differences between the $Z=0$ and $Z=1$ samples that affect outcomes, these differences would get bundled into the newly-hired teacher effect $\beta_{NT}$.



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where $\delta \equiv P(Z=0|D=1)$ denotes the proportion of school-grades that had one class in the previous year among school-grades with two classes. The second and third equalities follow from Assumptions 1 and 2, respectively.

Equation (3.7) makes clear that the RD estimand, $\tau_{SRD}$, combines the effect of decreasing class size and altering teacher quality,[15] except in the two special cases noted in Section 2: newly-hired teachers are the same quality as current teachers ($\beta_{NT}=0$) or there are no 'add a class' cases ($\delta=0$). If, as expected, the newly-hired teacher is of lower quality ($\beta_{NT}<0$), then the RD design underestimates the pure class size effect since teacher quality is higher in 'untreated' schools relative to 'treated' schools.

## 3.3 Separately Identifying Class Size and Teacher Quality Effects

To separately identify both the pure class size effect and the newly-hired teacher effect, I create two unique RD estimands: $\tau^r_{SRD}$ and $\tau^a_{SRD}$, with the superscripts indicating that the estimand is restricted to 'remove a class' and 'add a class' variation, respectively. Although both estimands operate at the same discontinuity as the usual RD estimand, ensuring that treatment status is random given Assumption 1, $\tau^r_{SRD}$ exploits variation from school-grades exiting treatment, while $\tau^a_{SRD}$ uses variation from school-grades entering treatment.

The two unique RD estimands, $\tau^r_{SRD}$ and $\tau^a_{SRD}$, are created by restricting the sample to those in which were treated last period (i.e, $Z=1$) and those that were not (i.e., $Z=0$), respectively. To ensure $\tau^r_{SRD}$ is identified solely from the sample where $Z=1$ (i.e., from school-grades exiting treatment), I restrict the estimand to school-grades whose enrollment in the previous year is above this year's effective class size threshold (i.e., $X_{sg,t-1}>0$). For such school-grades, those with current enrollment below the threshold remove a class and its associated teacher and thus exit treatment, while those with enrollment above the threshold do not remove class and thereby do not exit treatment. In both cases, the school-grade does not require a newly-hired teacher, indicating that there are no newly-hired teachers on either side of the discontinuity, although a teacher is removed among those school-grades that eliminated a class.

Similarly, the 'add a class' estimand, $\tau^a_{SRD}$, is restricted to the sample where $Z=0$ by limiting the data to school-grades whose prior year's enrollment lies below their effective class size cap this year (i.e., $X_{sg,t-1}<0$). In this case, treated school-grades add a new class, thereby entering treatment, and must hire a teacher because they have an insufficient number of teachers from the prior year. Therefore, since the discontinuity in outcomes exploited by

[15] In instrumental variable terminology, using an instrument based on a maximum class size rule to identify the effect of class size on student achievement leads to a violation of the exclusion restriction due to (unobserved) teacher quality being correlated with the instrument. Identification of both effects requires two instruments, as in Angrist and Lang (2004).



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$\tau^r_{SRD}$ requires no newly-hired teacher, $\tau^r_{SRD}$ identifies the pure class size effect. $\tau^a_{SRD}$, on the other hand, requires an additional teacher in the treated case and thus identifies the joint effect of smaller class sizes and newly-hired teachers.

To implement these RD designs, I break the forcing variable into two: the enrollment increase required to add a class, $X^a_{sgt}$, and the enrollment decline required to remove a class, $X^r_{sgt}$, as follows:

$$X_{sgt} = \begin{cases} X^a_{sgt} \text{ if } X_{sg,t-1} \leq 0 \\ X^r_{sgt} \text{ if } X_{sg,t-1} > 0\,. \end{cases} \tag{3.8}$$

Defining the forcing variable in this manner separates the sample into two cases: where the school-grade is on the verge of adding a class and where the school-grade is on the verge of removing a class. To give an example, suppose that school-grade enrollment in the prior year is 40 and the maximum class size rule is 32, so that $X_{sg,t-1} = 8$. For the school-grade to remove a class, it must have an enrollment loss of 8 students relative to the prior year. If enrollment drops by 7 students, the school-grade (just) does not achieve the enrollment decline necessary to remove a class and so $X^r_{sgt} = 1$, while a fall in enrollment by 8 students is (just) sufficient to remove a class, making $X^r_{sgt} = 0$. Hence, in the first case, the school-grade is treated (and does not have to hire a teacher), while in second case the school-grade is untreated. An analogous argument establishes that $X^a_{sgt}$ captures school-grades where school-grades are on the verge of adding a class, which entails hiring an additional teacher.

**Identification of the Class Size Effect ($\boldsymbol{\beta_{CS}}$):** The 'remove a class' estimator, $\tau_{SRD}$, is restricted to the sample where $Z = 1$ and therefore compares outcomes for those treated in both the current period *and* the pre-period ($\mathbb{E}[Y|D=1, Z=1, x^r_{sgt}=x]$) to those that are untreated in the current period but were treated in the pre-period ($\mathbb{E}[Y|D=0, Z=1, x^r_{sgt}=x]$ at $x=0$). This comparison yields the following result:

**Proposition 1** *Under Assumptions 1 and 2, $\tau^r_{SRD}$ identifies the pure class size effect.*

**Proof.**

$$\begin{aligned} \tau^r_{SRD} &= \mathbb{E}\left[Y|D=1, Z=1, X=0\right] - \mathbb{E}\left[Y|D=0, Z=1, X=0\right] \\ &= \mathbb{E}\left[Y(1,2)|Z=1, X=0\right] - \mathbb{E}\left[Y(0,1)|Z=1, X=0\right] \\ &= \mathbb{E}\left[Y(1,2) - Y(0,1)|Z=1, X=0\right] \\ &= \mathbb{E}\left[Y(1,2) - Y(0,1)|X=0\right] \equiv \beta_{CS}\,, \end{aligned} \tag{3.9}$$

where the second equality follows from Assumption 1 and the fourth equality relies on Assumption 2. ■

**Identifying the Newly-Hired Teacher Effect ($\boldsymbol{\beta_{NT}}$):** Similarly consider the RD



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estimand with its forcing variable defined as $X^a_{sgt}$, which is the school-grade enrollment change required to add a class. This estimand, denoted by $\tau^a_{SRD}$, does have an underlying discontinuity in newly-hired teachers, and thus we have:

$$\begin{aligned}
\tau^a_{SRD} &= \mathbb{E}\left[Y|D=1, Z=0, X=0\right] - \mathbb{E}\left[Y|D=0, Z=0, X=0\right] \\
&= \mathbb{E}\left[Y(1,2_N)|Z=0, X=0\right] - \mathbb{E}\left[Y(0,1)|Z=0, X=0\right] \\
&= \mathbb{E}\left[Y(1,2_N)|X=0\right] - \mathbb{E}\left[Y(0,1)|X=0\right] \\
&= \mathbb{E}\left[Y(1,2_N)|X=0\right] \underbrace{-\mathbb{E}\left[Y(1,2)|X=0\right] + \mathbb{E}\left[Y(1,2)|X=0\right]}_{=0} - \mathbb{E}\left[Y(0,1)|X=0\right] \\
&= \underbrace{\mathbb{E}\left[Y(1,2_N) - Y(1,2)|X=0\right]}_{=\beta_{NT}} + \underbrace{\mathbb{E}\left[Y(1,2) - Y(0,1)|X=0\right]}_{=\beta_{CS}} \\
&\equiv \beta_{NT} + \beta_{CS}\,,
\end{aligned} \tag{3.10}$$

where the second equality follows from Assumption 1 and the third equality follows from some rearrangement and the definition of $\beta_{CS}$ and $\beta_{NT}$ in Section 3.1.

I take the difference in $\tau^r_{SRD}$ from $\tau^a_{SRD}$ to capture the effect of crossing the 'adding a class' threshold that comes from the change in teacher characteristics. Formally, defining $\tau_{diff} \equiv \tau^a_{SRD} - \tau^r_{SRD}$ as the difference-in-discontinuity estimand (see Grembi, Nannicini, and Troiano (2016)), we have:

**Proposition 2** *Under Assumptions 1 and 2, $\tau_{diff}$ identifies the newly-hired teacher effect.*

**Proof.** This follows immediately from Equations (3.9) and (3.10). ■

Under Assumptions 1 and 2, $\tau^r_{SRD}$ obtains the pure effect of class size on student achievement while $\tau_{diff}$ captures the newly-hired teacher effect.

**Estimation:** While all estimands can be recovered using separate RD regressions, I estimate them simultaneously using the following regression:

$$\begin{aligned}
y_{sgt} &= \alpha_1 + \beta_1 D_{sgt} + \theta_1 \tilde{X}_{sgt} + \phi_1 \left(D_{sgt} * \tilde{X}_{sgt}\right) + \omega_1 Z_{sgt} \\
&\quad + Treat_{sgt} * \left[\alpha_2 + \tau_{diff} D_{sgt} + \theta_2 \tilde{X}_{sgt} + \phi_2 \left(D_{sgt} * \tilde{X}_{sgt}\right) + \omega_2 Z_{sgt}\right] + \epsilon_{sgt}, \\
&\quad for\ -b \leq \tilde{X}_{sgt} \leq b\,,\ where\ \tilde{X}_{sgt} = \begin{cases} X^r_{sgt} \text{ if } -b \leq X^r_{sgt} \leq b \\ X^a_{sgt} \text{ if } -b \leq X^a_{sgt} \leq b\,, \end{cases}
\end{aligned} \tag{3.11}$$

where $y_{sgt}$ is the average test score of school $s$ in grade $g$ at time $t$, $D_{sgt}$ is a treatment dummy ($D_{sgt} = 1$ if $\tilde{X}_{sgt} > 0$, and zero otherwise), $X^r_{sgt}$ and $X^a_{sgt}$ are the forcing variables




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defined in Equation 3.8,[16] $b$ is some bandwidth,[17] $Treat_{sgt}$ is a dummy variable indicating that the school is closer to removing (rather than adding) a class, and $Z_{sgt}$ is a set of school-grade-year controls. Estimates of both $\tau^a_{SRD}$ and $\tau^r_{SRD}$ are then derived from Equation 3.11: $\hat{\tau}^a_{SRD} \equiv \hat{\beta}_1$ and $\hat{\tau}^r_{SRD} \equiv \hat{\beta}_1 + \hat{\tau}_{diff}$.

## 3.4 Validity of Metholodogical Design

As in any RD design, the validity of the methodological design can be checked using standard RD tests.[18] However, there are potential concerns unique to my design involving teacher turnover and the non-random release of teachers. These concerns create potential discontinuities in the 'remove a class' case, violating Assumption 1, and generating a *downward* bias in the estimated pure class size effect. I alleviate these concerns by first providing a method to correct for teacher turnover directly, eliminating the downward bias in the pure class size effect induced by teacher turnover creating a discontinuity in the proportion of newly-hired teachers in the 'remove a class' case. Second, I show that non-random releasing of teachers generates a downward bias in the pure class size effect. The pure class size effect identified in this paper can thereby be considered a lower bound to the true class size effect, although – given the results – the downward bias does not appear to be substantial. Finally, I discuss the way that the 'pure class size effect' here also incorporates benefits due to student tracking and differential class sizes that may or may not be present in a randomized experiment, depending on whether schools can alter student assignment across classrooms.

**Teacher Turnover:** A complication is that teacher turnover causes $\tau^r_{SRD}$ to be underestimated as there exists a differential probability of hiring a teacher around the class size cap in the 'remove a class' case. This occurs because, in the treated case, a teacher retirement would, for example, force the school-grade to hire a teacher. However, a teacher retirement does not cause a teacher to be hired in the untreated case because the school-grade is able to avoid releasing the teacher whom it would normally release when it removes a class. I correct for the differential probability of a newly-hired teacher around the class size cap explicitly by adjusting the $\tau^r_{SRD}$ upward using the estimated newly-hired teacher effect. In cases where the newly-hired teacher effect is close to zero or, as in my setting, where the rate of teacher turnover is small, this correction has little effect on the estimated pure class size effect.

Let $\lambda$ be the probability that a teacher leaves in a given year. The number of teachers and

[16] Since $\tilde{X}_{sgt}$ is discrete and a school-grade with a zero value of $\tilde{X}_{sgt}$ is untreated, 0.5 is subtracted off $\tilde{X}_{sgt}$ so that zero is the midpoint between the first treated and untreated observations.

[17] The bandwidth should not exceed half the class size cap (sixteen) to avoid counting the same observation twice in a multiple class size cap setting.

[18] Specifically, testing for bunching around the class size cap and continuity in observables at the 'add a class' and 'remove a class' thresholds provide a sensible check of Assumption 1 (see Section 5.1).



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newly-hired teachers $T(D, Z, \lambda)$ are now a function of: (i) treatment status in the current period, $D$, (ii) treatment status in the prior period, $Z$, and (iii) the teacher turnover rate, $\lambda$. I now examine the discontinuity in the number of newly hired teachers under teacher turnover. (See Appendix D for a more detailed setup in this context and full derivations.)

In the 'remove a class' case, treated school-grades have a newly hired teacher with probability $\lambda$ and two newly-hired teachers with probability $\lambda^2$. Once a class is removed, however, the untreated school-grades only have a newly-hired teacher with probability $\lambda^2$ because they only have to find an additional teacher if *both* teachers leave since one teacher is released regardless. Assuming that two newly-hired teachers generate twice the effect of one newly-hired teacher, this implies that in addition to the class size effect there is a $\lambda \cdot \beta_{NT}$ discontinuity in outcomes around the 'remove a class' cap. If the newly-hired teacher effect is negative, then not accounting for teacher turnover will cause the pure class size effect to be underestimated.

Similarly, in the 'add a class' case, untreated school-grades have a newly hired teacher with probability $\lambda$ and no newly-hired teacher with probability $1 - \lambda$, while treated school-grades have one newly-hired teacher with probability $1-\lambda$ and have two newly-hired teachers with probability $\lambda$ since they must always add a teacher when they exceed the class size cap. Assuming that there is no interaction between teacher quality and class size,[19] the discontinuity in outcomes around the 'add a class' cap is thus $(1-\lambda)\cdot\beta_{NT} + \beta_{CS}$, rather than the $\beta_{NT} + \beta_{CS}$ discontinuity in outcomes in the absence of teacher turnover.

To correct for teacher turnover, I use the observed teacher turnover rate $\lambda$ to adjust the two estimands $\tau^r_{SRD}$ and $\tau^a_{SRD}$. Specifically, given the discussion above, equations (3.9) and (3.10) become:

$$\tau^r_{SRD} = \beta_{CS} + \lambda\beta_{NT} \tag{3.12}$$

$$\tau^a_{SRD} = \beta_{CS} + (1 - \lambda)\beta_{NT}\,. \tag{3.13}$$

Given a consistent estimate of $\lambda$,[20] equations (3.12) and (3.13) form a system of two equations with two unknown parameters, $\beta_{CS}$ and $\beta_{NT}$. Hence, given estimates of $\tau^r_{SRD}$, $\tau^a_{SRD}$ and $\lambda$, the pure class size effect and the newly-hired teacher effect are identified.

[19] Mueller (2013) suggests that there is a complementarity between teacher experience and class size. Under this case, there is a minor upward bias in the estimated class size effect, though this will have a negligible impact since this bias is multiplied by $\lambda^2$ and $\lambda$ is small. (See Appendix D for more details.)

[20] The researcher can estimate $\lambda$ either using an RD around the 'remove a class cap' or, under the assumption that $\lambda$ is not a function of treatment itself, as the rate of teacher turnover in the sample. In the context of New York City, it is unlikely that $\lambda$ would be a function of treatment itself because class sizes are not set until well after the start of the school year (see Section 4 for details). Thus, teachers cannot respond to treatment until several months into the school year, which is after retirement or departure decisions are made. Due to data limitations that I only observe teacher turnover at the school level, I calculate $\lambda$ as the rate of teacher turnover in the sample.


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Appendix D solves the system of two equations and provides full derivations of equations (3.12) and (3.13).

**Releasing of Teachers:** The random release of teachers is necessary for Propositions 1 and 2. In this subsection, I relax this assumption by drawing on the prior literature and institutional details from my application (see Section 4) to show that the $\tau^r_{SRD}$ and $\tau_{diff}$ estimands capture a lower bound for the pure class size effect and an upper bound for the newly-hired teacher effect. Intuitively, this arises since non-random teacher releases benefit untreated school-grades, which works against finding a pure class size effect.

In New York City, schools can choose which teacher to release when a class is removed, making teacher releases non-random. If schools have leeway to choose, we expect that they will release the lowest quality teacher. This is corroborated in the New York City data for the 2009-10 school year,[21] where the value-added of teachers who are released is about $0.03\sigma$ lower than the teachers who are not released. In other contexts, it is common for schools to be forced by union contracts to drop the least experienced teacher (as in New York City before 2005). Even here, we expect that, *on average*, lower quality teachers will be released since prior research has found that experience raises teacher effectiveness.[22] Thus, under non-random releases, the lower quality teacher is released in expectation, and so untreated school-grades perform better in the 'remove a class' case than under random teacher releases (i.e., $\mathbb{E}\left[Y(0,1)|Z=1,X=0\right] \geq \mathbb{E}\left[Y(0,1)|Z=0,X=0\right]$). It becomes clear that, in this case, $\tau^r_{SRD}$ captures a lower bound:

**Proposition 3** *$\tau^r_{SRD}$ identifies a lower bound to the pure class size effect.*

**Proof.**

$$\begin{aligned}\tau^r_{SRD} &= \mathbb{E}\left[Y|D=1,Z=1,X=0\right] - \mathbb{E}\left[Y|D=0,Z=1,X=0\right] \\ &= \mathbb{E}\left[Y(1,2)|Z=1,X=0\right] - \mathbb{E}\left[Y(0,1)|Z=1,X=0\right] \\ &= \mathbb{E}\left[Y(1,2)-Y(0,1)|Z=1,X=0\right] \\ &\leq \mathbb{E}\left[Y(1,2)-Y(0,1)|X=0\right] \equiv \beta_{CS}\,,\end{aligned} \tag{3.14}$$

where the second equality follows from Assumption 1 and the inequality follows from lower quality teachers being released when a class is removed, so that $\mathbb{E}\left[Y(0,1)|Z=1,X=0\right] \geq \mathbb{E}\left[Y(0,1)|X=0\right]$. ■

**Corollary 1** *$\tau_{diff}$ identifies an upper bound to the newly-hired teacher effect.*

[21] The teacher value-added data come from the New York Times release of New York City teacher value-added data. The data only cover the 2007-08 through 2009-10 school years.

[22] For instance, Hanushek and Rivkin (2010) find that early teacher experience is an important predictor of teacher quality.



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The corollary follows from an analogous argument to that in Proposition 3: Appendix D lays out the formal proof. Since we expect that the class size effect is positive while the newly-hired teacher effect is negative, non-random teacher releases act to attenuate the estimates in this paper toward zero, although given the effect sizes estimated below, it appears that this attenuation is small.

**Student Tracking and Differential Class Sizes:** The assumptions of no student tracking and no differential class sizes have been maintained throughout. Since schools can utilize student tracking and differential class sizes whenever there are two classes, but cannot when there is one class, both of these possible school responses are incorporated into the class size effect. In each case, we expect that these school responses raise the benefits of adding a class and thus reducing class sizes. While these responses do not occur in randomized experiments that randomly assign students and teachers to small or large classes (such as Project STAR), these responses are available to schools in randomized experiments that provide them with additional teachers (as in Duflo, Dupas, and Kremer (2015)) or whenever class sizes are reduced by a class size reduction policy.

Student tracking arises when students are grouped into classes by ability level, creating high- and low-ability classes.[23] Since having two classes rather than one allows schools to track students into different classes, the 'pure class size' effect in this paper incorporates such student tracking. Prior research, most notably Duflo, Dupas, and Kremer (2011),[24] finds that tracking raises student achievement, although some studies have found null results (Cohodes, forthcoming). Overall, it appears that tracking is likely to increase the 'pure class size' effect in this paper.

Differential class sizes comes from schools, rather than splitting classes such that each class has a class size of $\frac{N_t}{2}$, dividing students so that there are $\frac{N_t}{2}+\zeta$ students in class 1 and $\frac{N_t}{2}-\zeta$ students in class 2. So long as the class size effect is linear, this differential tracking will not affect the interpretation of the class size effect. The newly-hired teacher effect, however, will be decreased if administrators assign less students to classes taught by the newly-hired teacher. While such tactics are in line with models of classroom education (Lazear, 2001), empirically such strategic differential class size assignment appears muted (Hansen, 2014).

[23]Schools may also try to mix different ability students together, although Kalogrides and Loeb (2013) find that this type of tracking is rare.

[24]Duflo et al. (2011) implemented a randomized experiment in Kenya whereby treated schools tracked classes by ability, while control schools randomly allocated students to classes. They find that tracking has a positive effect on essentially all students. In another context, Card and Giuliano (2016) find that tracking students into gifted classrooms raised student achievement.




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# 4 Background and Data

New York City (NYC) features a typical U.S. education structure for elementary schools (which is the focus of the empirical analysis), whereby students generally have one teacher that teaches all subjects for that year. Teachers predominantly instruct the same grade year after year, and so students typically receive a different teacher every year. The remainder of this section describes New York City's class size rule and its implementation. In addition, I discuss the pool of teachers that school-grades hire from as these teachers form the basis for the 'newly-hired' teacher effect identified in this paper. I also describe the data for my application of the methodology using New York City's class size cap.

## 4.1 New York City's Class Size Cap

New York City's (NYC) maximum class size rule stems from its long history of bargaining with the union representing its teachers, the United Federation of Teachers (UTF). The class size rule in NYC dates back to 1963 when, for the first time in the United States, class size limits were written into a teacher-Board of Education contract. The class size limits used in this paper are from the 2007-09 collective bargaining agreement. This contract, which remained in force through the 2013-14 school year due to a failure to reach any new agreement, stipulates class size limits of 25 for kindergarten, 32 in elementary school grades, 33 in middle school grades, and 34 in high school grades.

While the contract specifies class size caps for each grade, school-grades often start the school year with class sizes well above the allowable limits. The New York City Department of Education (NYC DOE) has the first ten school days to ensure class sizes are within their contractual limits. Afterwards, enforcement of the class size cap comes through grievances that the UTF files against the NYC DOE for any class that exceeds its contractual cap.[25] Arbitrators decide those grievances on a case-by-case basis, but almost always mandate the NYC DOE to abide by the collective bargaining agreement and reduce class sizes below the class size cap.[26] However, it often takes several months before the arbitration case is decided, indicating that school-grades often exceed the class size cap for at least part of the school year.

[25]From 2009-13, the UTF launched about 4,000-6,500 grievances per year. Prior to 2009, the author cannot find evidence of the UTF filing grievances – likely leading to the observed minimal discontinuity in average class size at the class size cap prior to 2009. The UTF launched 6,749 grievances in 2009, 4,461 grievances in 2010 , (approx.) 4000 grievances in 2011, 6,133 grievances in 2012 and 6,313 grievances in 2013.

[26]There are some exemptions to the maximum class size rule, including a lack of classroom space (cannot be used for two consecutive years), scheduling concerns, creation of half-classes, and a specialized instruction exemption. For the main sample in the paper, only 74 school-grades had average class sizes exceed 32 students in a given year (about 1% of the sample).



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## 4.2 The Absent Teacher Reserve

Since this paper estimates the effect of a newly-hired teacher on student achievement, institutional details about the available teacher pool are relevant in order to understand the group of teachers producing the newly-hired teacher effect. The available teacher pool in NYC is unique due to the presence of the Absent Teacher Reserve (ATR) during the period of study. While the ATR creates a rather context-specific newly-hired teacher effect, it provides sufficient identifying variation to separate out the pure class size and newly-hired teacher effects.

The presence of the ATR stems from a 2005 agreement between the Bloomberg administration and the UTF giving principals the right to choose whom to hire and retain.[27] Prior to the agreement, hiring was dictated by seniority, with principals being forced to hire the most senior teacher – principals were even compelled to release a teacher when a more senior teacher applied for that position. The agreement gave principals significant leeway and led them to release senior teachers they deemed inadequate. These teachers had tenure, however, and thus could not be fired from the NYC DOE. With no principal willing to hire these teachers, the NYC DOE formed the ATR for those without assignments but with guaranteed salaries. These unwanted teachers, unsurprisingly, appear to be low in quality: of all the teachers in the ATR, twenty-five percent committed wrongdoing, twenty-five percent are rated 'unsatisfactory' (only 3% of NYC teachers are rated unsatisfactory), forty percent are from schools closed or downsized due to poor performance, and ten percent taught subjects no longer offered (Lestch, 2014).

Due to the high cost of paying salaries to teachers not performing teaching duties,[28] the NYC DOE attempted to eliminate the ATR. In 2009, it encouraged ATR hiring by implementing a teacher hiring freeze and providing a subsidy equal to the difference in salary between an ATR teacher and a teacher with no experience to any school hiring an ATR teacher. The UFT and the NYC DOE continued curtailing the ATR with a June 2011 deal. The deal ensured that every long-term absence or leave is filled by an ATR teacher and ATR teachers are interviewed for vacancies and assigned as day-to-day substitutes for one-week intervals in different schools. The deal was reaffirmed for the 2012-13 school year by a union-initiated grievance to increase enforcement. Hence, NYC's pool of available teachers

[27] The ATR differs from New York City's so-called 'rubber rooms' for teachers who could not legally teach due to pending misconduct claims. ATR teachers are fully licensed to teach, but are unable to find a position at any school. The 'rubber rooms' were all but eliminated in 2010 due to an agreement between the NYC DOE and the UTF which expedited misconduct claims. Many former 'rubber room' teachers entered the ATR.

[28] Even accounting for the fact ATR teachers often work as substitutes, the NYC DOE estimated the ATR cost to be $105 million in the 2013 fiscal year.



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is somewhat context-specific during the period of study; yet, this institutional feature helps provide the substantial divergence in student outcomes between the 'add a class' and 'remove a class' cases that is the focus of the econometric methodology.

## 4.3 Data

The data are drawn from the NYC DOE, which provides information about class sizes, school-grade enrollment and student demographics for the school years 2009-10 through 2013-14. In addition, the NYC DOE provides test score results from the New York State English Language Arts & Mathematics Tests for grades 3-6 for any school-grade with more than ten student test takers. These high-stake tests are unique to New York and are used for student retention decisions.[29] The math test includes questions relating to number sense and operations, geometry, and measurement, while the English test includes multiple-choice and short-response questions based on a reading and listening section. All public-school students are tested, with exemptions only for medical reasons or severe disabilities. Some students are granted special accommodations (additional time, translation services, etc.) if they have limited English proficiency or a moderate disability. In the 2012-13 school year, the test was adapted to comply with Common Core standards, making year-to-year comparisons difficult. I thus standardize each school-grade-year's score by the (enrollment-weighted) mean and standard deviation of scores in my sample for that grade-year.

Several sample restrictions are made in the analysis. I use only school-grades that do not have a gifted and talented (G&T) or integrated co-teaching (ICT) class in their school-grade due to bunching around the class size cap (see Section 5.1 for an explanation). I also restrict my analysis to grades 3-6 in an elementary school so that my sample consists of school-grades with self-contained classrooms, clear outcomes measures, and straightforward class size rules.[30] Further, following Hoxby (2000), I analyze only the effects around class size cap discontinuities that generate a decline in class size of greater than twenty percent, which restricts my analysis to the first three maximum class size caps.[31] Finally, I restrict to school-grades with non-missing enrollment data from the prior year, because otherwise it is unclear how the forcing variables should be defined.[32]

[29]Students in third, fifth, and seventh grades must score at level 2 or above (out of 4) on both the math and English tests to advance to the next grade without attending summer school.

[30]Test scores are only available for grades 3 and above. The class size cap for grades 4 through 8 in a middle school supposedly depends on Title I status (class size rules of 30 and 33 students for Title I and non-Title I schools, respectively), although the enforcement of these rules are unclear. High school grades (9-12) are not taught in self-contained classrooms.

[31]The next class size cap only generates a nineteen percent decline in class size if it is followed. In practice, this restriction eliminates less than fifteen percent of the observations.

[32]This sample restriction eliminates a further three percent of observations.



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Table 1 presents average school-grade characteristics for the 2009-10 through 2013-14 school years for grades 3-6 within an elementary school. Results are presented for all relevant school-grades within NYC and for the restricted sample used in this analysis. Looking at column (2), which contains the restricted sample, we see that almost all students are tested and that the NYC DOE is a majority-minority school district with nearly four-fifths of its student population being either black or Hispanic. Additionally, the average class size is twenty-four students.[33]

Table 1 also presents school-grade characteristics around the threshold to 'add' or 'remove' a class as described in the econometric framework. Comparing school-grades in the 'add a class' sample versus the 'remove a class' sample, some differences are apparent. Column (5) performs a statistical test of whether the differences between the two cases are statistically significant. School-grades in the 'remove a class' sample have a higher fraction of black students with a correspondingly smaller fraction of white students than the 'add a class' sample, though the difference is only about three percent. In addition, the 'remove a class' sample has substantially smaller class sizes than the 'add a class' sample. Also, school-grades in the 'remove a class' sample are generally declining in enrollment, while school-grades in the 'add a class' sample are generally increasing in enrollment – a mechanical by-product of how the forcing variables are defined because school-grades induced to add (remove) a class due to the maximum class size rule must have increasing (declining) enrollment.

# 5 Results

In this section, I discuss the validity of my design in the context of New York City's maximum class size rule, then provide graphical evidence followed by formal regression estimates. Finally, I examine the robustness of the results. Throughout the analysis, standard errors are clustered at the school-grade enrollment level following Lee and Card (2008). I conclude the section by comparing my class size estimates with results from the most notable randomized experiment on class size: Project STAR.

## 5.1 Validity of the Empirical Design

The validity of the empirical methodology described in Section 3 depends on continuity of potential outcomes (Assumption 1). Testing for continuity of both covariates and the density

[33]The class size data I use are from the mid-February report in each school year. However, at that time some class size grievances are winding through arbitration and so some non-compliant classrooms may still be arbitrated back into compliance. Additionally, some data in February may not get updated from the September report.



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of the forcing variable at the threshold provides a sensible check of the validity of Assumption 1. I present evidence of continuity of density and covariates at the maximum class size cap simultaneously for the three RD estimators described in this paper: the standard RD estimator $\tau_{SRD}$, the 'add a class' RD estimator $\tau^a_{SRD}$, and the 'remove a class' RD estimator $\tau^r_{SRD}$. Testing for continuity of potential outcomes must be carried out for all estimators separately: testing solely for continuity of outcomes over the standard RD estimator $\tau_{SRD}$ is insufficient as, in these asymmetric contexts, a continuous density could be observed even when underlying manipulation around the 'add a class' and 'remove a class' thresholds took place.[34]

Testing for a continuous density in school-grade enrollment around the thresholds assuages concerns that school-grades manipulate their enrollment to sort in or out of treatment,[35] which would invalidate continuity of potential outcomes. A visual inspection of whether school-grades manipulate their enrollment is provided in Figure 1, which plots the distribution of school-grades by enrollment. Visually, there is excess density around the threshold in Figure 1(a) which represents all school-grades. However, Ballou (1999) demonstrated that the class size cap in New York City causes schools to shift students between gifted and talented (G&T) or Integrated Co-Teaching (ICT) classes and regular classes.[36] In Figure 1(b), I remove these school-grades from the analysis and plot the distribution of school-grades without ICT and G&T programs. In this restricted sample, the excess density noted in Figure 1(a) disappears. Therefore, there is no evidence of manipulation of the forcing variable for the restricted sample. Figures 1(c) and 1(d) plot the distribution by school-grade enrollment for the 'add a class' and 'remove a class' samples; a visual inspection confirms that there is no underlying bunching around these thresholds. A formal test of continuity in the density around the threshold (McCrary, 2008) confirms our visual analysis:

[34]For example, consider a case where school-grades do not wish to add a class due to the need to hire a teacher, but also do not wish to remove a class due to parental pressure. In this case, school-grades bunch to the left of the discontinuity in the 'add a class' case and to the right of the discontinuity in the 'remove a class' case. Accordingly, there may be no bunching around the discontinuity for the standard RD estimator, but substantial bunching around the discontinuity in both the 'add a class' and 'remove a class' cases. Therefore, separate density tests for each case are required.

[35]Manipulation can be carried out by either school administrators or parents. Administrators may manipulate enrollment to avoid adding a new class, while parents may respond to the school-grade's treatment status by enrolling their child in a different school. These manipulations are difficult in New York City: school attendance zones limit administrators' ability to influence enrollment and a school-grade's treatment status is not observed until well into the school year (see Section 4), which – because enrollment decisions are made before school starts – restricts parents' ability to place their child in another school.

[36]ICT school-grade classes are for students with disabilities who receive teaching in a general education classroom with the support of a special education teacher to assist them. While class size limits exist for non-ICT students, the ICT students in the class do not count toward the limit as they have their own teacher. Due to this, a school can designate non-ICT students to ICT to remain below the class size cap. A similar story occurs for G&T school-grades. Around sixty percent of school-grades are ICT and G&T.



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the null hypothesis of continuity at the cutoff is rejected for all school-grades. However, once the sample is restricted, the null hypothesis of continuity at the cutoff cannot be rejected for either the standard RD design or the 'add a class' and 'remove a class' samples.

Similar to randomized experiments, continuity of potential outcomes can also be explored by ensuring continuity in observable characteristics at the thresholds. Figure 2 and Figures A.1 and A.2 graph mean school-grade covariates by the forcing variable for the standard RD, 'add a class' and 'remove a class' designs, respectively. These covariates include race, disability status, English learner status, enrollment, enrollment change and the test-taking rate. For all covariates, smooth distributions around the cutoffs are observed. More formally, point estimates from an RD regression with a bandwidth of ten are reported above the figures. As expected, at five percent significance, none of the covariates exhibit a statistically significant discontinuity at the cutoff. Further evidence of continuity of covariates is provided in Table A.1, which uses a seemingly unrelated regression to test the null hypothesis that all covariates are jointly continuous at each threshold. Given the p-values, the null hypothesis cannot be rejected for any design.

## 5.2 Estimates

To identify $\tau^a_{SRD}$ and $\tau^r_{SRD}$, the class size rule must induce school-grades to reduce (increase) average class sizes by adding (removing) a class when enrollment is above (below) the class size cap. If the class size rule is followed, there will be a drop in average class size just to the right of the discontinuity for both the 'add' and 'remove a class' thresholds.[37] Evidence of this 'first-stage' is provided in Figure 3, which plots the average class size by normalized school-grade enrollment for all three designs. As predicted, there is a large change in average class size at each threshold. Above each figure, I carry out a formal test of the statistical significance of the drop in average class size at the discontinuity. The coefficient is significant at the one percent level and signifies that crossing the class size cap leads to a drop in average class size of three to four students – a 12-14 percent decline – in all three designs. In addition, Figure A.3 plots actual class size by predicted class size (see Angrist and Lavy (1999)) for all three designs to provide a sense of the degree of compliance around the class size caps.

Figures 4 and 5 plot the reduced-form relationship between the forcing variables and normalized test scores for math and English, respectively. Once the class size cap is crossed and school-grades become treated, both math and English test scores fall in the 'add a class' case, but increase in the 'remove a class' case. The point estimates above the figures

[37]This is because the class size rule induces school-grades just to the right of the class size cap to add another class in the 'add a class' design and to not remove a class in the 'remove a class' design.




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correspond to the estimates of $\tau^a_{SRD}$ and $\tau^r_{SRD}$ provided in columns (1) and (3) of Table 2. Table 2 also supplies difference-in-discontinuity estimates ($\tau_{diff}$), which are the difference between $\tau^a_{SRD}$ and $\tau^r_{SRD}$.

The point estimates for $\tau^a_{SRD}$, $\tau^r_{SRD}$ and $\tau_{diff}$ in Table 2 are estimated using Equation 3.11 and are local intent-to-treat (ITT) estimates. Local treatment-on-the-treated (TOT) estimates are obtained by dividing local ITT estimates by the discontinuity in the probability that a school-grade has an additional class at the threshold, which in this instance is about one-third. Consequently, local TOT estimates are three times the magnitude of the local ITT estimates, although local TOT estimates are generated from a class size reduction threefold larger than in Figure 3. If the class size effect is linear,[38] the TOT and ITT interpretations are identical: a 3-4 student decline in average class size is associated with the change in tests scores given by the local ITT estimate. Throughout the paper, I report local ITT estimates.

Columns (2) and (4) of Table 2 provide estimates corresponding to Figures 4 and 5 with the inclusion of covariates. Since the covariates are continuous through the discontinuity, point estimates are similar, though standard errors are smaller; consequently, I focus the discussion on results with covariates. The standard RD estimator, $\tau_{SRD}$, suggests a positive effect on student outcomes, however, this estimator combines both the effects of smaller classes from the 'remove a class' case and the joint effect of smaller classes and a newly-hired teacher from the 'add a class' case. Decomposing the standard RD estimator, large differences between the two cases are observed. Estimates from the 'add a class' case estimator, $\tau^a_{SRD}$, suggest that school-grades (just) induced to add a new class have a small decline in student achievement for both math and English, although this decline is not statistically significant. Since $\tau^a_{SRD}$ is estimated solely from school-grades adding a class relative to school-grades not adding a class, its estimates are interpreted as the effect of adding a new class on student achievement. The 'remove a class' case estimates from $\tau^r_{SRD}$, on the other hand, show a significant increase in student achievement among school-grades not removing a class. For math, this effect is statistically significant at the five percent level and the difference in these two cases, represented by $\tau_{diff}$, is at the statistically significant at the ten percent level. Due to teacher turnover, these estimates do not directly correspond to parameter estimates, but the magnitude of the class size effect roughly equals $\tau^r_{SRD}$, while the newly-hired teacher effect approximately equates to $\tau_{diff}$.

Parameter estimates under teacher turnover are obtained by solving the system of equations formed by equations (D.4) and (D.5). Since the underlying first stages in these two

[38] Under non-linearity, the local ITT and local TOT estimates differ as the change in student achievement generated by a twelve-student decline in class size is not three times the magnitude of a four-student decline in class size.



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equations vary, the estimates could be scaled to account for the different first stages: yet, because the first stages do not statistically differ and first-stage reporting is not entirely reliable (see footnote 33), I do not scale the estimates here. In practice, scaling does not influence the results much.[39] Table 3 reports the parameter estimates. The pure class size parameter indicates that a class size reduction of approximately four students yields an increase in math and English scores of $0.21\sigma$ and $0.13\sigma$, respectively. The results are significant at the five percent level for math. Overall, much of the positive impact of smaller classes is offset by the newly-hired teacher, who reduces test scores at the school-grade level by $0.26\sigma$ and $0.19\sigma$ in math and English, respectively. These estimates are noisier than estimates of the pure class size parameter; nonetheless, the newly-hired teacher effect is statistically significant at the ten percent level for math.[40]

## 5.3 Sensitivity Analysis

I use a bandwidth of ten throughout this paper. The sensitivity of the estimated parameters with respect to the bandwidth is tested in Figure A.4, which plots parameter estimates for both math and English for $b \in [5, 15]$. Although English results are somewhat sensitive to bandwidth choice, the parameter estimates for math do not vary substantially with the bandwidth and, as expected, precision improves as the bandwidth increases.

Estimates in the paper are generated with a linear functional form allowing for different functions on either side of the cutoff. The functional form between student outcomes and the forcing variables around the discontinuity may not be linear, however. Following Gelman and Imbens (2019), Table A.2 reports the parameter estimates for two additional functional forms: quadratic and a triangular kernel. For English, results with the quadratic functional form lie below those using linear or triangular kernel functional forms, though standard errors are high. For math, results are very consistent across the three functional forms, although the class size parameter loses statistical significance under the quadratic functional form (p-value$=0.16$) due to an increase in standard errors.

The underlying assumption of the regression-discontinuity design is that outcomes would be continuous through the threshold in the absence of the maximum class size rule (Assumption 1). While indirect tests of this assumption are provided in Section 5.1, no direct test is available. Nevertheless, smoothness of outcomes over placebo maximum class size rules are indicative of smoothness at the true threshold in the absence of the maximum class size rule. Figure A.5 reports the parameter estimates over a range of placebo maximum class

[39] Results with scaling are available upon request.

[40] The fact the newly-hired teacher effect is noisier than the class size effect is to be expected as there is likely to be substantial variation in the quality of newly-hire teachers, leading to noisier estimates of the LATE.



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size rules. In general, the plurality of maximum class size rules yield an estimate near zero. The sole exception is the maximum class size rules of 29-31 students, which show effect sizes near those reported for the true maximum class size rule of 32. However, it is likely that these rules are picking up some of the discontinuity in test scores at the true threshold and thus outcomes appear smooth except near the policy discontinuity.

## 5.4 Comparison of Results to Project STAR

I now compare my class size estimates with results from the most notable randomized experiment on class size: Project STAR. To make effect sizes comparable, I use the experimental data provided by Achilles et al. (2008) to create Project STAR estimates that are analogous to those in this paper. First, I restrict the Project STAR data to grades 2 and 3 to better match the grade levels in my analysis. Second, I collapse the Project STAR student-level data to a school-grade-treatment level to mimic the school-grade level analysis used in this paper. Despite these changes, I find estimates in line with the experimental class size estimates of Project STAR from Finn and Achilles (1990) and Krueger (1999).[41]

Table 4 compares the class size effects in Table 3 to those from Project STAR. I estimate a class size effect of $0.21\sigma$ and $0.13\sigma$ for math and English, respectively, whereas Project STAR reports effect sizes of $0.37\sigma$ and $0.47\sigma$ for math and English, respectively. However, since the effects are estimated from different reductions in class size, with Project STAR's first-stage being about twice the size, I scale the effect sizes from Project STAR linearly to the underlying first-stage in this paper.[42] Once I account for the difference in first-stages, class size effects in this paper are similar to those from Project STAR. The experimental class size effect from Project STAR is slightly lower in math ($0.17\sigma$ compared to $0.21\sigma$), but larger in English ($0.21\sigma$ compared to $0.13\sigma$), although the difference is not statistically different.

The results here show that in the context of New York City the RD design is able to deliver class size effects in line with Finn and Achilles (1990) and Krueger (1999) once the confounding effect of teacher quality is accounted for. Given that RD designs generally find a class size effect below those from experimental designs, the methodology proposed here has the potential to reconcile two strands of empirical research. Angrist and Lavy (1999), for instance, find class size effects a bit below or at the low end of the range of those found

[41]See Table 5 of Finn and Achilles (1990) and page 514 of Krueger who report Project STAR yielded effect sizes in the distribution among pupils in the 0.2-0.28$\sigma$ range for math and English. Finn and Achilles (1990) further report effect sizes of $0.64\sigma$ and $0.66\sigma$ in the distribution of class means for grades K-1 in math and English scores, respectively. My analysis of effect sizes in the distribution of school-grade-treatment means most closely aligns with effect sizes in the distribution of class means in Finn and Achilles (1990).

[42]The first-stage decline in class sizes are, however, from a different base level: Project STAR has a base level of around 23 students, while in NYC the base level of students is around 26.



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in Project STAR:[43] These estimates are likely to increase once the potential discontinuity in teacher quality is accounted for, aligning results from Angrist and Lavy (1999)'s RD design to those from the experimental design of Finn and Achilles (1990) and Krueger (1999).

# 6 Conclusion

This paper has proposed a novel extension of the RD design that allows researchers to disentangle individual treatment effects from a discontinuity incorporating multiple treatment components. It does so by exploiting an asymmetry between entities entering versus exiting treatment that is encountered in a variety of empirical settings.

I apply my approach to estimate the pure effect of class size, where a perennial debate among many policymakers and economists concerns the merits of reducing class sizes. One important source of contention has been the way the empirical class size literature has bifurcated. Effect sizes from the experimental class size literature lead proponents to conclude that "[r]ecent research confirms what parents have always known – children learn better in small classes with good teachers, and kids who start out in smaller classes do better right through their high school graduation" (Clinton, 1999), while opponents of class size reduction note that quasi-experimental evidence provides "little reason to believe that smaller classes systematically yield higher student achievement" (Hanushek 1999, page 153). As noted above, however, these literatures measure two different effects: a pure class size effect and the combined effect of smaller class sizes and newly-hired teachers.

My analysis reconciles the two strands of empirical research by decomposing the effects of class size reduction into its two underlying components: changes in class size and changes in teacher quality. The results in this paper indicate that class size reduction can substantially improve student achievement if teacher quality is held constant. When teacher quality is not held constant, however, class size reduction is unlikely to generate large improvements in student achievement. The results here do not rely on general equilibrium shifts in teacher quality, in contrast to compelling prior work by Jepsen and Rivkin (2009) that investigated a large-scale class size reduction program; instead, they rely solely on newly-hired teachers differing from current teachers. The key to the efficacy of class size reduction therefore hinges on the ability of policymakers to hold teacher quality fixed, perhaps by enhancing teacher training in conjunction with class size reductions.

The innovative design used in this analysis can be used more generally to identify unique treatment effects in a discontinuity composed of multiple components. The general conditions

[43]Specifically, Angrist and Lavy (1999) find class size effects at the low end of the Project STAR range for fifth grade, but find that class size effects for fourth grade are about half those for fifth grade.



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are that the researcher has panel data, there is year-to-year treatment variation induced by entities receiving shocks to their forcing variable, and where one treatment component only occurs when entities enter or exit treatment.

As one possible general application, the methodology can be used to capture underlying preference or technological parameters. For instance, one can easily adapt the methodology to compare the treatment effect among individuals who had received treatment the year prior (i.e., remain treated) to those who had not (i.e., entered treatment). Under a technology that features dynamic complemetarities, the treatment effect should be larger in the former case, allowing for the underlying technological parameter to be identified – something I am exploring further in related work (see Gilraine (2018)).



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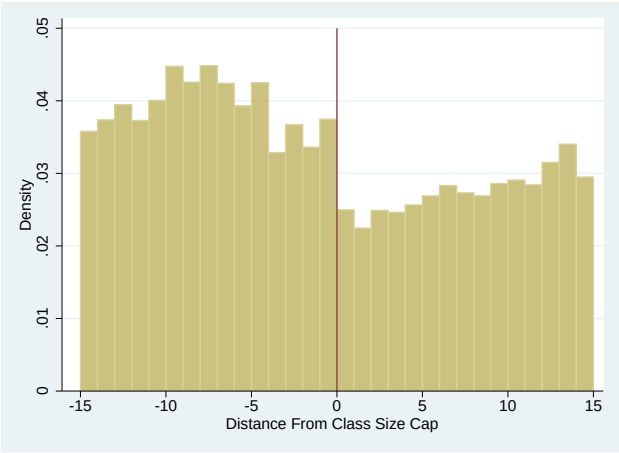
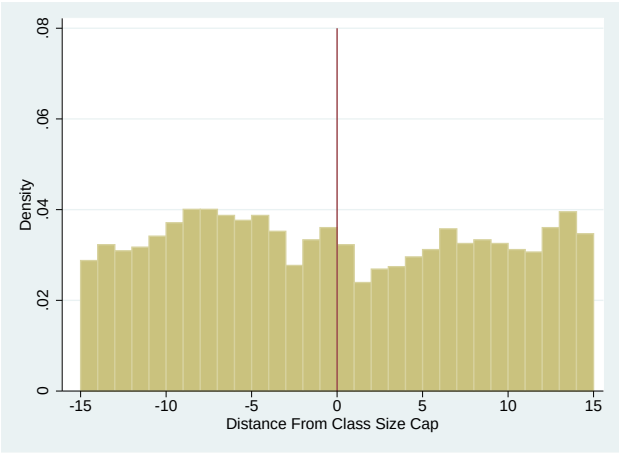
Figure 1: Histograms

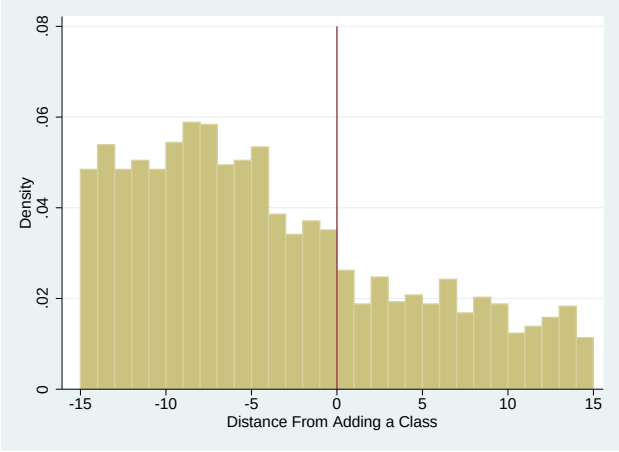
(a) All School-grades
McCrary p-value: 0.00



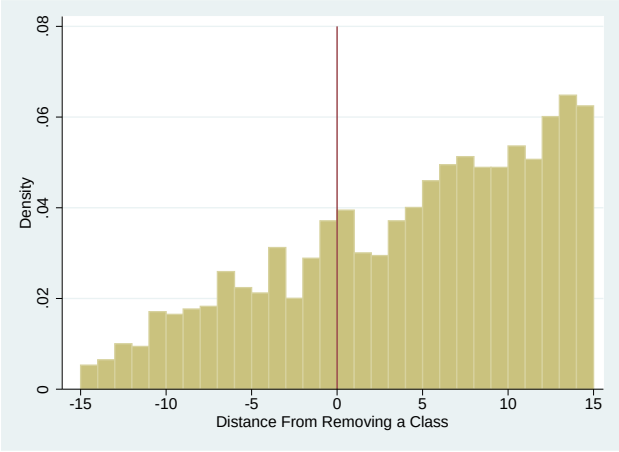

(b) Restricted Sample
McCrary p-value: 0.26




(c) Add a Class
McCrary p-value: 0.12




(d) Remove a Class
McCrary p-value: 0.97




Notes: Figures 1(a), 1(b), 1(c) and 1(d) are based on 11,934, 3,719, 2,022 and 1,697 observations, respectively. All histograms have a bin-size of one. P-values are reported for a McCrary (2008) test using a bin-size of one and a bandwidth of five.



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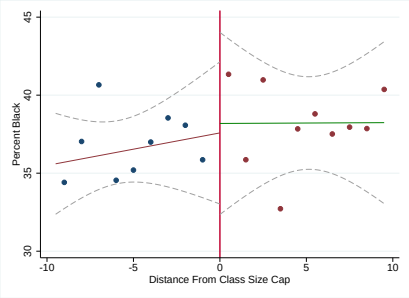
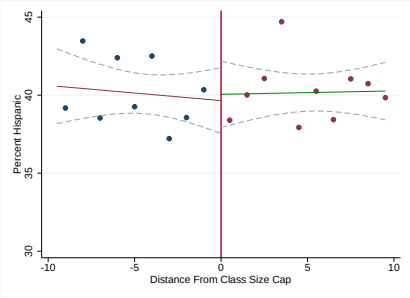
Figure 2: Covariates (Standard RD)

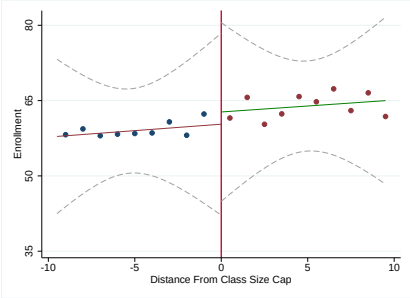
(a) Percent Black
RD Estimate: 0.61 (3.59)



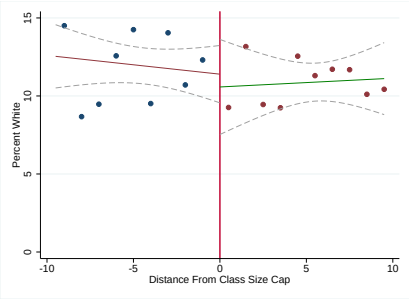

(b) Percent Hispanic
RD Estimate: 0.40 (1.45)



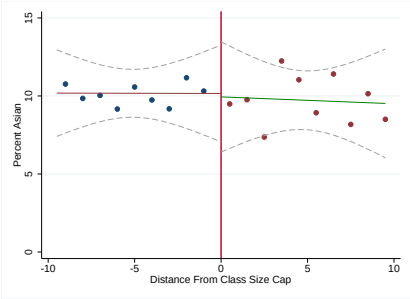

(c) Enrollment
RD Estimate: 2.44 (12.32)



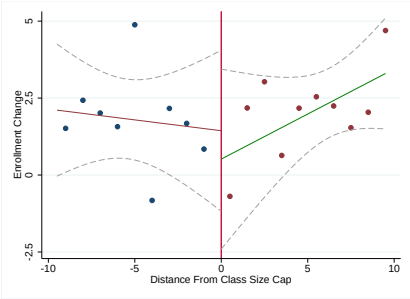

(d) Percent White
RD Estimate: -0.82 (1.73)



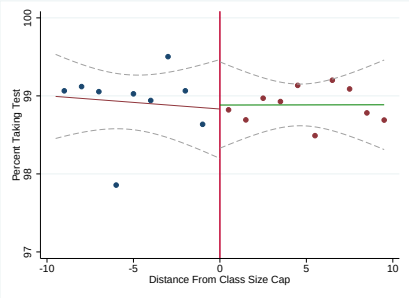

(e) Percent Asian
RD Estimate: -0.22 (2.28)



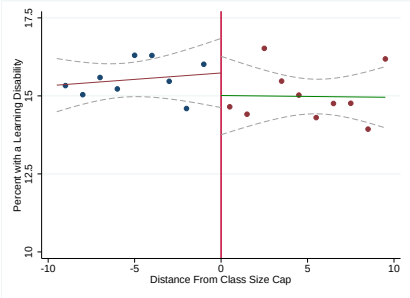

(f) Enrollment Change
RD Estimate: -0.92 (1.90)



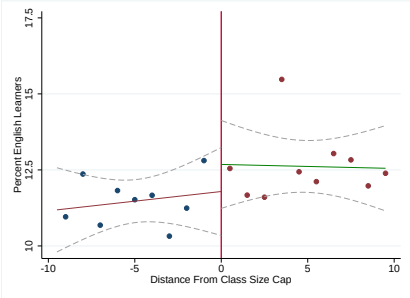

(g) Percent Taking Test
RD Estimate: 0.05 (0.41)




(h) Percent Students With a Disability
RD Estimate: -0.72 (0.81)




(i) Percent English Learners
RD Estimate: 0.89 (0.99)




Notes: Figures are based on 2,489 observations. Each RD estimate is from a separate local linear regression allowing for different functions on either side of the threshold. The bandwidth used is ten. Dashed lines represent 95% confidence intervals with standard errors clustered on school-grade enrollment following Lee and Card (2008). Point estimates correspond to those in Panel A of Table A.1. Significance levels: *** 1 percent; ** 5 percent; * 10 percent.



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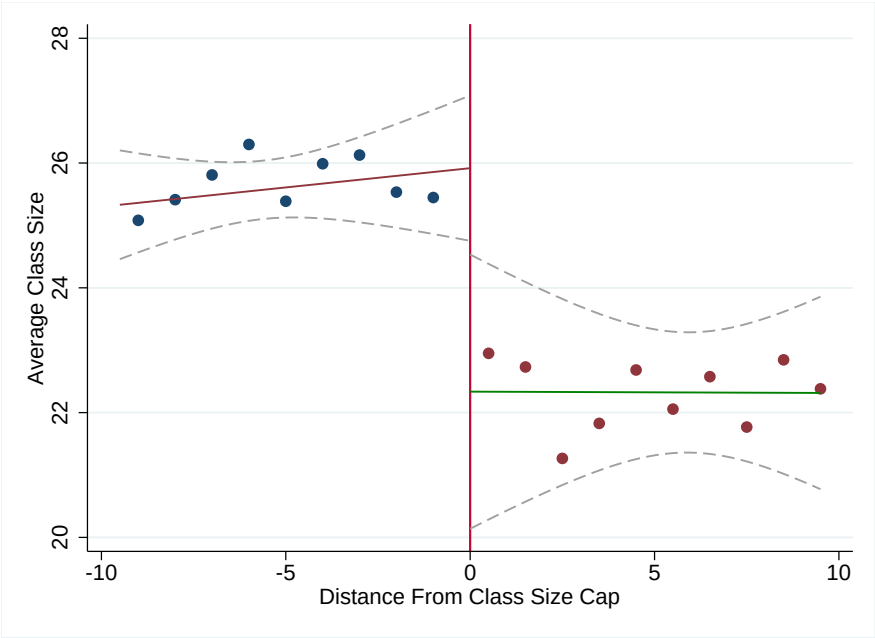
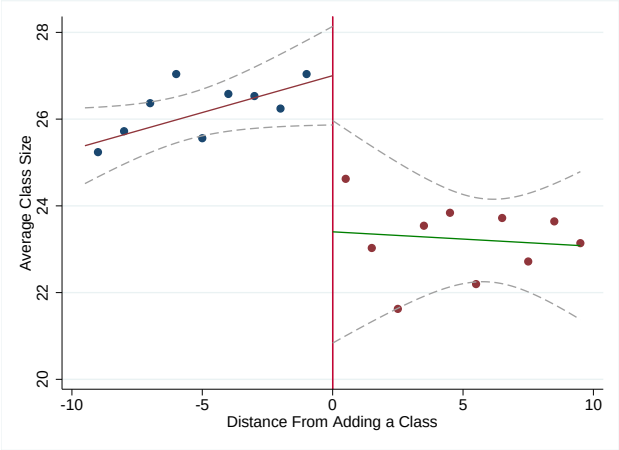
Figure 3: First-Stage (Average Class Size)

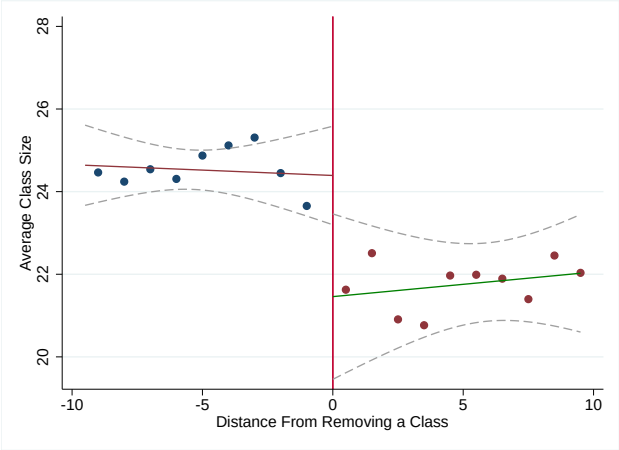
(a) Class Size - All
RD Estimate: -3.58*** (1.21)




(b) Class Size - Add a Class
RD Estimate: -3.60*** (1.36)




(c) Class Size - Remove a Class
RD Estimate: -2.93*** (1.13)




Notes: Figure 3(a) is based on 2,489 observations. Figures 3(b) and 3(c) are based on 1,371 and 1,118 observations, respectively. The y-axis represents average class size at the school-grade-year level as of mid-February. Each RD estimate is from a separate local linear regression allowing for different functions on either side of the threshold. The local linear regression does not include controls and uses a bandwidth of ten. Dashed lines represent 95% confidence intervals with standard errors clustered on school-grade enrollment following Lee and Card (2008). Significance levels: *** 1 percent; ** 5 percent; * 10 percent.



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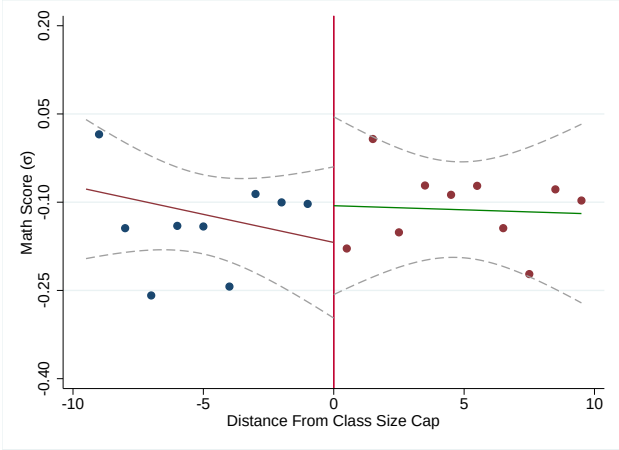
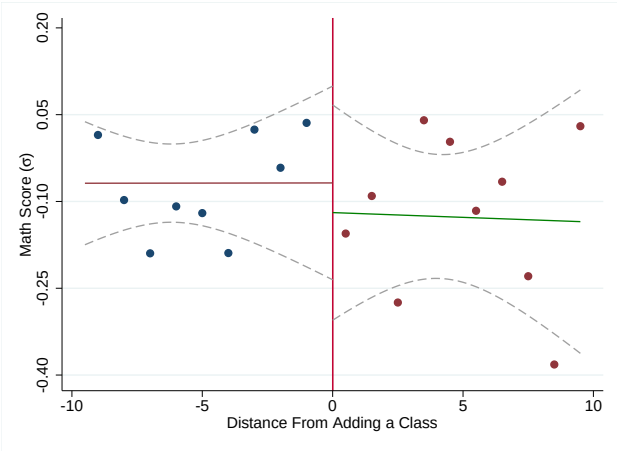
Figure 4: Reduced-Form - Math Score ($\sigma$)

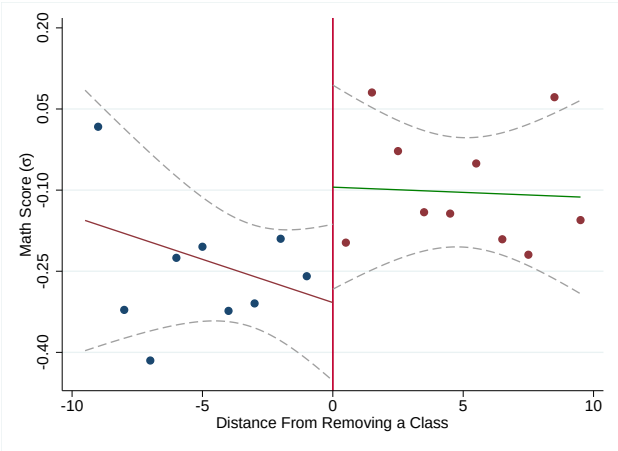
(a) Math Score - All
RD Estimate: 0.062 (0.096)




(b) Math Score - Add A Class
RD Estimate: -0.051 (0.121)




(c) Math Score - Remove A Class
RD Estimate: 0.213* (0.115)




Notes: Figure 4(a) is based on 2,489 observations. Figures 4(b) and 4(c) are based on 1,371 and 1,118 observations, respectively. Each RD estimate is from a separate local linear regression allowing for different functions on either side of the threshold. The local linear regression does not include controls and uses a bandwidth of ten. Dashed lines represent 95% confidence intervals with standard errors clustered on school-grade enrollment following Lee and Card (2008). Point estimates correspond to those in column (1) of Table 2. Significance levels: *** 1 percent; ** 5 percent; * 10 percent.



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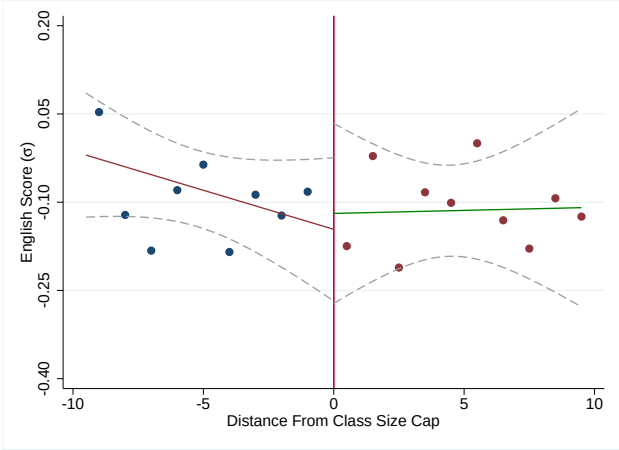
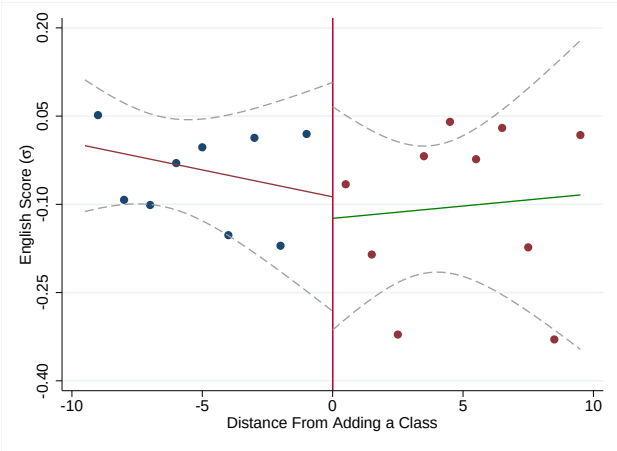
Figure 5: Reduced-Form - English Score ($\sigma$)

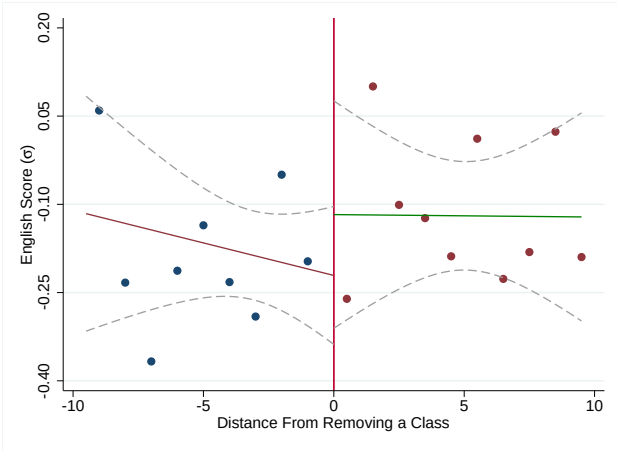
(a) English Score - All
RD Estimate: 0.027 (0.095)




(b) English Score - Add A Class
RD Estimate: -0.037 (0.132)




(c) English Score - Remove A Class
RD Estimate: 0.103 (0.110)




Notes: Figure 5(a) is based on 2,489 observations. Figures 5(b) and 5(c) are based on 1,371 and 1,118 observations, respectively. Each RD estimate is from a separate local linear regression allowing for different functions on either side of the threshold. The local linear regression does not include controls and uses a bandwidth of ten. Dashed lines represent 95% confidence intervals with standard errors clustered on school-grade enrollment following Lee and Card (2008). Point estimates correspond to those in column (3) of Table 2. Significance levels: *** 1 percent; ** 5 percent; * 10 percent.



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Table 1: Summary Statistics

| | All School-grades[1] (1) | Restricted Sample[2] (2) | 'Add a Class' Sample (+/- 10)[2,3] (3) | 'Remove a Class' Sample (+/- 10)[2,3] (4) | Add=Remove (p-value) (5) |
|---|---|---|---|---|---|
| *Mean (S.D.) Among School-grade Characteristics* | | | | | |
| Average Class Size | 24.98 | 24.00 | 25.20 | 22.77 | 0.00 |
| | (4.31) | (4.45) | (4.82) | (4.70) | |
| Math Score ($\sigma$)tnote4 | -0.11 | -0.11 | -0.09 | -0.15 | 0.11 |
| | (1.04) | (1.03) | (1.03) | (1.04) | |
| English Score ($\sigma$)tnote1 | -0.09 | -0.08 | -0.06 | -0.14 | 0.06 |
| | (1.04) | (1.05) | (1.03) | (1.04) | |
| % Hispanic | 40.19 | 39.83 | 40.31 | 39.97 | 0.77 |
| | (27.07) | (27.84) | (27.24) | (28.48) | |
| % Black | 31.53 | 37.59 | 35.58 | 39.37 | 0.01 |
| | (30.05) | (31.45) | (30.46) | (31.86) | |
| % White | 14.46 | 11.45 | 12.75 | 9.96 | 0.00 |
| | (22.03) | (20.11) | (21.50) | (19.05) | |
| % Asian | 12.65 | 10.05 | 10.26 | 9.60 | 0.32 |
| | (18.87) | (17.61) | (17.52) | (18.00) | |
| % of Students With Disability | 17.18 | 15.11 | 15.11 | 15.48 | 0.21 |
| | (7.97) | (7.98) | (7.98) | (7.85) | |
| % English Learners | 12.90 | 11.89 | 11.35 | 12.77 | 0.00 |
| | (11.63) | (10.88) | (9.95) | (11.87) | |
| % Taking Test | 99.59 | 98.97 | 98.85 | 98.97 | 0.42 |
| | (2.55) | (3.32) | (3.78) | (3.82) | |
| Enrollment | 80.32 | 61.55 | 62.45 | 59.78 | 0.24 |
| | (58.20) | (24.27) | (23.96) | (24.67) | |
| Enrollment Change | -2.01 | 1.83 | 10.79 | -8.98 | 0.00 |
| | (20.85) | (14.11) | (10.63) | (9.95) | |
| Observations (School-Grade-Year) | 12,893 | 3,951 | 1,371 | 1,118 | 2,489 |

[1] All school-grades include all general education school-grades in grades 3-6 within an elementary school with non-missing data for the prior year.
[2] The sample is restricted to school-grades without a gifted & talented or an integrated co-teaching program due to bunching and to school-grades with an enrollment less than 113 students as described in Section 4.3.
[3] The 'Sample (+/- 10)' term indicates the sample is restricted to a school-grade enrollment around the class size cap with a bandwidth of ten. This is the sample employed for all major results in this paper.
[4] Test scores are standardized at the grade-year level using school-grade enrollment weights. As bigger schools generally perform better than smaller schools, without enrollment weights the mean standardized test score is negative in the data.

Notes: Data cover school years 2009-10 through 2013-14. Sample is limited to observations with non-missing test data. Test results are not reported for school-grades with 10 or fewer test taking students. Column (5) indicates the p-value of a hypothesis test for equality between the ' Add a Class Sample (+/- 10)' and ' Remove a Class Sample (+/- 10)' with standards errors clustered by school-grade enrollment.



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Table 2: Regression Discontinuity Estimates (Reduced-Form)

| | *Math Scores ($\sigma$)* | | *English Scores ($\sigma$)* | |
|---|---|---|---|---|
| | No Covariates | Covariates | No Covariates | Covariates |
| | (1) | (2) | (3) | (4) |
| *A. RD Estimator* | | | | |
| $\tau_{SRD}$ | 0.062 | 0.080 | 0.027 | 0.048 |
| | (0.096) | (0.049) | (0.095) | (0.044) |
| *B. 'Add a Class' Estimator* | | | | |
| $\tau^a_{SRD}$ | -0.051 | -0.017 | -0.037 | -0.036 |
| | (0.121) | (0.062) | (0.132) | (0.076) |
| *C. 'Remove a Class' Estimator* | | | | |
| $\tau^r_{SRD}$ | 0.213* | 0.179** | 0.103 | 0.107 |
| | (0.115) | (0.086) | (0.110) | (0.070) |
| *D. Diff-in-disc Estimator* | | | | |
| $\tau_{diff}$ | -0.265* | -0.196* | -0.140 | -0.143 |
| | (0.147) | (0.113) | (0.164) | (0.121) |
| Observations | 2,489 | 2,489 | 2,489 | 2,489 |

Notes: Estimates report the effects of smaller class sizes. For $\tau^a_{SRD}$, there are 1,371 observations, while for $\tau^r_{SRD}$, there are 1,118 observations. Effect sizes are in standard deviations in the distribution of school-grade means. Sample is restricted to grades 3-6 within elementary schools without a gifted & talented or an integrated co-teaching program. $\tau_{SRD}$ is estimated using a separate RD regression, while all other estimators are estimated from Equation 3.11. The bandwidth used is ten. Covariates include controls for enrollment, enrollment change, student ethnicity, student disability status, percent English learners, test-taking rate, grade-by-year dummies, and NYC borough dummies. Standard errors are clustered on school-grade enrollment following Lee and Card (2008). ***,** and * denote significance at the 1%, 5% and 10% levels, respectively.



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Table 3: Parameter Estimates (Reduced-Form)

| | *Math Scores ($\sigma$)* | | *English Scores ($\sigma$)* | |
|---|---|---|---|---|
| | No Covariates | Covariates | No Covariates | Covariates |
| | (1) | (2) | (3) | (4) |
| $\lambda$ | 0.117*** | 0.117*** | 0.117*** | 0.117*** |
| (Teacher Turnover) | (0.005) | (0.005) | (0.005) | (0.005) |
| $\beta_{CS}$ | 0.254* | 0.209** | 0.125 | 0.128 |
| (Pure Class Size Effect) | (0.130) | (0.101) | (0.127) | (0.085) |
| $\beta_{NT}$ | -0.345* | -0.256* | -0.183 | -0.186 |
| (Newly-hired Teacher Effect) | (0.192) | (0.147) | (0.214) | (0.157) |
| Observations | 2,489 | 2,489 | 2,489 | 2,489 |

Notes: Estimates report the effects of smaller class sizes. Sample is restricted to grades 3-6 within elementary schools without a gifted & talented or an integrated co-teaching program. Besides teacher turnover, which is in percent, effect sizes are in standard deviations in the distribution of school-grade means. The bandwidth used is ten. Covariates include controls for enrollment, enrollment change, student ethnicity, student disability status, percent English learners, test-taking rate, grade-by-year dummies, and NYC borough dummies. Standard errors are calculated using the delta method and are clustered on school-grade enrollment following Lee and Card (2008). ***,** and * denote significance at the 1%, 5% and 10% levels, respectively.



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Table 4: Pure Class Size Effects Relative to Project STAR

| | *Math Scores (σ)* | | *English Scores (σ)* | |
|---|---|---|---|---|
| | This Study | Project STAR | This Study | Project STAR |
| | (1) | (2) | (3) | (4) |
| First-Stage (# of students) | -3.58*** | -7.90*** | -3.58*** | -7.89*** |
| | (1.21) | (0.19) | (1.21) | (0.20) |
| Reduced-Form | 0.209** | 0.370*** | 0.128 | 0.471*** |
| | (0.101) | (0.088) | (0.085) | (0.095) |
| *Scaling Project STAR's Estimates* | | | | |
| First-Stage Correction | 0.209** | 0.168*** | 0.128 | 0.214** |
| (Scaled to NYC first-stage) | (0.101) | (0.069) | (0.085) | (0.084) |
| Test of Equality | 0.66 | | 0.35 | |
| (p-value) | | | | |
| Observations | 2,489 | 290 | 2,489 | 288 |

Notes: Estimates report the effects of smaller class sizes. Parameter estimates for columns (1) and (3) are from columns (2) and (4) of Table 3 and are measured in standard deviations in the distribution of school-grade means. Effect sizes for columns (2) and (4) are in standard deviations in the distribution of school-grade-treatment means. Results for Project STAR use the experimental data available from Achilles et al. (2008) and are calculated using a restricted sample of students with valid grade 2 and 3 test scores. All Project STAR classes with the teacher aide treatment arm are omitted. No covariates are included in the Project STAR results and standard errors are clustered at the school level. Standard errors for the scaled estimates are calculated using the delta method. The test of equality tests the null hypothesis that the class size effect identified in this paper is the same as the class size effect identified in Project STAR. ***,** and * denote significance at the 1%, 5% and 10% levels, respectively.



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